Artificial Intelligence and Healthcare Economics

Evidence, Value, Governance, and Sustainability

Digital Resource Companion

Worked Examples, Supplemental Resources, Reference Guides, and Computational Materials

Muthana Zouri

Carmen Marinela Cumpăt

V1.0 | 2026

Comprehensive printable and editable edition aligned with the revised five-part, fifteen-chapter structure

Worked examples are synthetic unless explicitly stated.

Table of Contents

About This Companion

The **Digital Resource Companion** is the operational extension of *Artificial Intelligence and Healthcare Economics: Evidence, Value, Governance, and Sustainability*. The book presents the stable scholarly argument and the integrated decision architecture. This companion provides chapter-aligned materials that readers can adapt, complete, test, archive, and update without turning the book into a collection of rapidly changing forms.

The earlier printable companion organized institutional records into five broad resource sets. That version established useful common records for decision specification, economics, governance, procurement, lifecycle control, and circular value. The revised edition retains those functions but reorganizes them around the current five-Part, fifteen-chapter structure. Each chapter now has a worked example, supplemental records, reference resources, and links to the corresponding editable and computational materials.

The companion is intended for institutional analysis, education, research design, documentation, and deliberation. It is not a validated clinical instrument and does not constitute legal, regulatory, procurement, cybersecurity, accreditation, quality-certification, financial, or health technology assessment advice. Local evidence, lawful authority, professional judgement, multidisciplinary review, and accountable approval remain necessary.

Resource Architecture

Resource layer	Purpose	Best use
Worked examples	Demonstrate the complete path from problem and comparator through assumptions, analysis, uncertainty, interpretation, and conditional recommendation.	Read before adapting a template or building a local model.
Supplemental resources	Provide editable specifications, checklists, registers, matrices, protocols, monitoring plans, and decision records.	Use to document the local institutional process and preserve an auditable record.
Reference resources	Clarify terminology, methods, formulas, laws, standards, reporting guidance, appraisal tools, frameworks, and limitations.	Use to select the right instrument for the question and to avoid treating status or name recognition as evidence.
Core and specialist editable files	Supply DOCX/ODT, XLSX/ODS, CSV/JSON, Markdown, process-flow, and graphic variants.	Select the format that best fits the task while retaining the stable resource identifier and version.
Python analytics	Provide scripts, notebooks, structured synthetic data, tests, and outputs for quantitative tasks.	Use when reproducibility and computational analysis materially improve the decision.

The layers are designed to be used together. A worked example demonstrates the reasoning; a supplemental resource records the local work; a reference resource clarifies the status and limitations of the methods or instruments; and an editable or computational file supports implementation. No single file replaces the decision dossier.

Repository Navigation and Current Release

This release retains the chapter-aligned worked examples, supplemental records, reference resources, and computational materials while adding several cross-cutting ways to locate, combine, and exchange them. The repository remains the canonical operational source; the printable companion explains what is available, why it is included, and how the resource layers should be used together.

Master Decision Lifecycle and Decision Stages

A ten-stage Master Institutional Decision Lifecycle links problem definition, readiness, evidence, economics, modelling, governance, procurement, implementation, circular value, and renewal or retirement. Readers can browse by chapter when they already know the subject area or by decision stage when a project spans several chapters. Each stage identifies recommended preceding records, primary and supporting resources, a typical output, and a decision gate. The sequence is iterative rather than linear: validation findings, procurement constraints, incidents, material changes, or new alternatives may require a return to an earlier stage.

Process Maps and Browser-Editable Forms

The GitHub Pages layer exposes the master lifecycle, 15 chapter process maps, resource-level process maps, browser-editable versions of the 24 canonical core resources and seven cross-cutting reference guides, a searchable catalog, and interactive synthetic worked examples. Browser forms are drafting and learning aids. They do not establish evidence quality, compliance, readiness, or approval, and their exports should be transferred into a governed institutional record before operational use.

Python Analytics

A dedicated Python Analytics browser supports selection among 39 reproducible scripts and notebooks. Modules cover data quality, validation, calibration, decision utility, costing, budget impact, simulation, uncertainty, value of information, forecasting, optimization, post-deployment monitoring, and circularity measurement. Each module links to synthetic inputs, spreadsheet templates, example outputs, and related worked cases. Method choice should follow the decision structure rather than the availability of code, and consequential results require verification, transparent assumptions, version control, and an archived static decision output.

Consolidated Offline Bundles

Two release-level bundles support committees, workshops, and teams working offline. The consolidated forms volume provides the 24 canonical core records in DOCX and ODT. The consolidated workbook provides an index and one worksheet per resource in XLSX and ODS. These are convenience editions: the individual resource folder remains the canonical source, and users should verify the resource ID and version before reusing an older completed record.

Machine-Readable Records and Catalogs

Each canonical core resource and the Chapter 11 local-validation resource includes a blank JSON record, a synthetic worked JSON record, and a JSON Schema. Synchronized JSON catalogs are also provided for the master resource catalog, Python modules, worked cases, resource relationships, and figures. These files support digital forms, software integration, automated validation, and future APIs, but they do not replace institutional governance, evidence review, or approval.

How to Use the Companion

1. **Begin with the institutional decision.** Define the population, setting, accountable decision-maker, intended and excluded uses, comparator, time horizon, and consequence of error before selecting a file.
2. **Read the aligned book chapter.** The book remains the controlling source for conceptual claims, chapter structure, and the interpretation of the author-developed framework package.
3. **Review the worked example.** Identify the pathway from problem and comparator to evidence, cost, outcome, uncertainty, safeguards, and recommendation.
4. **Select the smallest sufficient resource set.** Avoid completing every form merely because it exists. Choose the records needed for the consequence, uncertainty, scale, and institutional dependence of the decision.
5. **Create a governed local copy.** Preserve the original resource identifier and release version, record the local owner, and work in an approved institutional location rather than the public repository.
6. **Replace synthetic assumptions.** Use current, authorized, versioned evidence. Mark unknowns explicitly and do not represent missing evidence as a neutral or average value.
7. **Complete multidisciplinary review.** Clinical, operational, patient, technical, economic, quality, privacy, cybersecurity, procurement, legal, and governance input should be proportionate to the decision.
8. **Separate observations, modelled estimates, value judgements, and contractual claims.** This distinction improves traceability and prevents unsupported certainty.
9. **End with an accountable decision.** Record the selected and rejected options, conditions, indicators, thresholds, owners, review date, trigger events, and exit route.

10. **Maintain the record after deployment.** Update the dossier after incidents, material changes, new evidence, price changes, altered workflows, renewal, restriction, replacement, or retirement.

Common Institutional Record

The following fields apply across the chapter resources and are presented once here to avoid repeating the same generic template in every chapter.

Field	Minimum entry
Decision and authority	Decision to be made; accountable sponsor; decision owner; approval, escalation, and stopping authority.
Population and setting	Included and excluded populations, sites, services, users, and time period.
Problem, baseline, and comparator	Current pathway, burden, variation, unmet need, and all realistic AI and non-AI alternatives.
Intervention boundary	Data, model or method, interface, people, workflow, capacity, implementation, monitoring, vendor support, and exit.
Evidence and version	Sources, dates, exact system and software version, applicability, confidence, gaps, assumptions, and evidence cut-off.
Costs and consequences	Perspective, budget holder, resource use, outcomes, workload, distribution, opportunity cost, and residual value where relevant.
Mandatory safeguards	Patient safety, lawful data use, privacy, cybersecurity, equity, accessibility, meaningful oversight, continuity, and contestability.
Decision conditions	Proceed, redesign, restrict, pilot, defer, replace, or retire criteria; milestones; contractual duties; and review date.
Monitoring and closure	Indicators, thresholds, triggers, responsible owners, material-change route, renewal decision, transition, data closure, and lessons.

Standard Decision Statement

> For the specified decision problem and comparator set, the institution should **[proceed / proceed with conditions / redesign / pilot / restrict / defer / replace / retire]**. The decision applies to **[population, setting, system and version, and evidence period]**, is owned by **[accountable authority]**, and will be reviewed on **[date]** or earlier if **[trigger events]** occur. The evidence, assumptions, unresolved uncertainty, mandatory safeguards, monitoring thresholds, material-change route, and exit arrangements are recorded in the accompanying dossier.

Reader Pathways

The reader-facing navigation begins with five goals: learn the approach, make an institutional decision, use a template or workbook, run an analysis, and teach or facilitate a case. Each route begins with a small curated set and reveals advanced source files only when they are needed.

Resources are classified as introductory, applied, advanced, or technical source. The Web Companion shows core records, worked cases, the common case-analysis method, and reference guides by default. Supplemental specialist tools, Python, LaTeX, schemas, metadata, and process source remain available through the Advanced view.

A separate Reader and Practitioner Edition provides the guided entry files, pathways, consolidated forms and workbooks, worked examples, reference guides, analytics catalog, printable companion, and offline web layer without exposing the complete technical source tree. The full GitHub repository remains the canonical source for maintenance, reproducibility, integration, and publication.

Reader group	Suggested route	Principal use
Boards, executives, and senior managers	Chapters 1–3, 5–9, and 12–15	Problem definition, portfolio priorities, evidence, lifecycle cost, procurement,

Reader group	Suggested route	Principal use
		governance, renewal, and sustainable value.
Clinical, nursing, allied-health, quality, and patient-safety leaders	Chapters 3–5 and 10–12	Human oversight, workflow, quality control, patient safety, incident response, and continuing assurance.
Health economists, HTA specialists, finance teams, payers, and commissioners	Chapters 1 and 6–9, followed by 11 and 15	Costing, pricing, ROI, cost-effectiveness, budget impact, affordability, uncertainty, and circularity-adjusted value.
Digital health, informatics, data, and implementation teams	Chapters 2–4, 8, 11, and 12	Data readiness, interoperability, validation, modelling, implementation, monitoring, and reproducibility.
Ethics, legal, privacy, cybersecurity, and institutional-assurance teams	Chapters 2, 3, and 9–12	Lawful use, rights, accountability, risk, audit, accessibility, contestability, and stopping authority.
Procurement, supply chain, biomedical engineering, asset management, and sustainability teams	Chapters 6, 9, and 12–15	Contracts, supportability, maintenance, repair, refurbishment, residual value, take-back, and disposition.
Researchers, educators, students, and professional learners	Begin with Chapters 1, 7, 8, and 10–12, then use the relevant case	Research design, appraisal, causal evaluation, modelling, governance analysis, workshops, and reproducible exercises.

Current Book and Companion Structure

Part	Title	Aligned chapters
Part I	Managerial and Economic Foundations	Chapter 1: AI as a Health-System Intervention; Chapter 2: Institutional Readiness and Digital Foundations; Chapter 3: Human–AI Decision Support and Managerial Action
Part II	Organizational Performance, Workforce, and Quality	Chapter 4: Productivity, Workforce Redesign, and Implementation; Chapter 5: Quality, Accreditation, and Organizational Learning
Part III	Economic Evaluation, Decision Modelling, and Procurement	Chapter 6: Lifecycle Cost, Pricing, and Benefit Realization; Chapter 7: Economic Evaluation, Affordability, and HTA; Chapter 8: Decision Modelling and Simulation Under Uncertainty; Chapter 9: Problem-First Procurement and Vendor Governance
Part IV	Governance, Evidence, and Lifecycle Decision-Making	Chapter 10: Governance, Accountability, and Institutional Risk; Chapter 11: Evidence, Validation, and Real-World Evaluation; Chapter 12: Decision Architecture and Lifecycle Governance
Part V	Circular Economy and Sustainable Value	Chapter 13: Circular Health Systems and Resource Stewardship; Chapter 14: Circular Procurement and Asset Stewardship; Chapter 15: Circularity-Adjusted Evaluation and Sustainable Value

File Formats and Repository Structure

Format	Typical purpose	Selection guidance
DOCX / ODT	Narrative records, committee papers, protocols, assurance notes, and local guidance.	Use when explanation, review comments, approvals, and versioned prose are central.
XLSX / ODS	Registers, scoring tables, cost logs, repeated records, scenarios, and dashboards.	Use when rows are updated over time or calculations and filters are needed.
CSV / JSON	Structured exchange, audit trails, schemas, and reproducible data workflows.	Use for controlled interchange and machine-readable records; keep a human-readable decision record alongside them.

Format	Typical purpose	Selection guidance
Markdown / Mermaid	Repository documentation, process descriptions, and editable flow diagrams.	Use for version-controlled guidance and transparent process maps.
SVG / PNG	Scalable and raster process visuals.	Use SVG for editable publication graphics and PNG for broad compatibility.
Python / Jupyter	Reproducible validation, costing, simulation, uncertainty analysis, monitoring, and circularity calculations.	Use only within an approved analytic environment with versioned inputs, code, tests, and outputs.
LaTeX	Publication-quality worked examples and printable companion material.	Use when stable mathematical notation, long tables, and formal typesetting are required.

A typical repository path is:

```
ai-healthcare-economics-companion/
├── README.md, GETTING_STARTED.md, BOOK_STRUCTURE.md, and RESOURCE_INDEX.md
├── catalogs/
│   ├── MASTER_RESOURCE_CATALOG.xlsx / .ods / .csv / .json
│   ├── WORKED_CASE_CATALOG.* and PYTHON_MODULE_CATALOG.*
│   ├── RESOURCE_RELATIONSHIP_MAP.* and FIGURE_CATALOG.json
├── process/ and docs/
│   ├── master lifecycle, decision stages, process maps, forms, and browser tools
├── bundles/
│   ├── consolidated core forms (DOCX/ODT) and workbooks (XLSX/ODS)
├── crosswalks/
│   ├── current legacy-resource mapping
├── chapters/01/ ... chapters/15/
│   ├── core, supplemental, worked-case, Python, flow, and LaTeX resources
├── latex/
└── publications/printable_companion/
```

The stable resource identifier, not the file path alone, should be recorded in institutional work. Folders and filenames may change between releases; the resource identifier, version, title, and release date provide the traceable citation.

Cross-Cutting Reference Guides

The cross-cutting guides address questions that recur across chapters and are especially likely to change as law, standards, reporting guidance, software practice, and institutional policy evolve.

ID	Guide	Purpose	Primary alignment
AIHE-RG01	Framework Status and Selection Guide	Distinguish legal instruments, standards, guidance, reporting guidelines, appraisal tools, implementation frameworks, and author-developed decision aids; select them according to function and authority.	Chapter 12; aligned 2, 4, 5, 7, 9-12
AIHE-RG02	Reporting and Appraisal Tool Selection Guide	Match evidence type and study stage to current reporting and appraisal instruments without treating checklist completion as proof of validity, effectiveness, or regulatory conformity.	Chapter 11; aligned 7, 8, 11
AIHE-RG03	Implementation Framework Selection Guide	Select CFIR, NASSS-CAT, RE-AIM/PRISM, TAM/UTAUT, and mixed-method approaches according to the implementation question rather than name recognition.	Chapter 4; aligned 4, 11, 12
AIHE-RG04	Regulatory, Standards, and Policy Update Register	Maintain date-stamped records for European, Romanian, and global	Chapter 10; aligned 2, 5, 7, 9, 10, 12-15

ID	Guide	Purpose	Primary alignment
		legal, policy, standards, reimbursement, and guidance sources whose scope or implementation status may change.	
AIHE-RG05	Scenario Planning and Real-Options Worksheet	Examine uncertainty through alternative futures, thresholds, staged commitments, learning options, contract flexibility, and explicit exit criteria.	Chapter 8; aligned 1, 6-9, 12, 15
AIHE-RG06	Diagnostic and Predictive Evidence Appraisal Checklist	Appraise reference standards, index tests, data partitions, calibration, diagnostic accuracy, bias, subgroup performance, clinical utility, and downstream resource consequences.	Chapter 11; aligned 3, 7, 8, 11
AIHE-RG07	Common AI Adoption Failures and Corrective Actions	Identify recurring technology-led, economic, workflow, evidence, governance, procurement, and lifecycle failures and record preventive controls.	Chapter 1; aligned 1-12

A guide organizes selection and documentation; it does not certify compliance, methodological quality, safety, effectiveness, cost-effectiveness, or readiness.

Version Control, Data Protection, and Citation

- Download or clone a versioned release and retain the unmodified source copy.
- Record the resource identifier, title, version, release date, date accessed, and local modification date.
- Maintain a change log identifying the field or formula changed, the reason, the owner, the evidence source, and whether reapproval is required.
- Record the exact model, software, data, interface, prompt, threshold, and workflow version when they affect the decision.
- Do not enter identifiable patient information, credentials, private keys, confidential contracts, or restricted institutional data into public or uncontrolled copies.
- Store completed resources according to institutional retention, access, audit, and incident-response requirements.
- Cite both the book and the specific companion release used. A local adaptation should be described as an adaptation rather than represented as the unchanged published resource.
- GitHub supports the active working repository; Zenodo or another archival repository should hold citable, immutable releases. Use the DOI and repository URL shown on the release page.

Status Taxonomy for External and Author-Developed Instruments

- **Law or regulation:** establishes binding obligations within its jurisdiction and scope.
- **Standard or management system:** structures repeatable practices or requirements but does not replace legal interpretation or local evidence.
- **Reporting guideline:** improves transparency of a report; completion does not prove methodological validity or effectiveness.
- **Appraisal tool:** supports critical review of bias and applicability; it does not make the institutional decision.
- **Implementation framework or theory:** organizes determinants, processes, outcomes, context, or acceptance questions.
- **Data or interoperability specification:** supports exchange or analysis but does not establish semantic consistency, data fitness, or readiness by itself.

- **Author-developed framework or process:** a proposed decision aid requiring explicit status, versioning, limitations, and independent evaluation before standardized use.

Select each instrument for a defined question, document why it was chosen, and verify its current edition and jurisdictional status before use.

Part I: Managerial and Economic Foundations

This Part establishes the managerial and economic problem, the institutional foundations required for action, and the design of human–AI decision support.

Chapter 1: AI as a Health-System Intervention

Chapter Orientation

Chapter 1 establishes the companion's governing unit of analysis: the complete health-system intervention rather than the model alone. The resources help a team define the service problem, compare AI and non-AI alternatives, identify the causal path from information to action and value, and preserve the ability to change or stop the decision.

Use this suite at project intake, portfolio prioritization, and before market engagement. It is especially useful when a proposal begins with a product claim rather than a documented problem, or when different departments are using inconsistent assumptions about scope, comparator, outcomes, ownership, and opportunity cost.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Emergency-Triage Intervention Portfolio	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Intervention Specification; Comparator and Opportunity-Cost Record; Decision Conditions and Review Record	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Intervention bundle, Comparator, Public value	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	1 case-analysis method, 2 core, 1 optional reference, 2 optional teaching, 1 reference guide, 3 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Emergency-Triage Intervention Portfolio

Purpose and Status

A regional public hospital is considering an AI-supported emergency-triage service after repeated crowding, prolonged time to first clinical assessment, and concern about delayed recognition of deteriorating patients. The proposal must be assessed as a complete intervention rather than as a prediction model.

Decision Problem and Comparators

Decision problem. Should the hospital pilot an AI-supported triage pathway, strengthen the current pathway through staffing and protocol redesign, or adopt a lower-complexity rules-based alternative?

Table: Comparators for the Chapter 1 worked example

Option	Comparator or strategy
1	Current triage with existing staffing
2	Additional triage staffing during peak periods
3	Registration and streaming redesign
4	A conventional rules-based early-warning score
5	AI-supported prioritisation with human confirmation

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$\text{Incremental value hypothesis: } \Delta V = f(\Delta O, \Delta C, \Delta W, \Delta R)$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 1 worked example

Parameter	Base value	Source or interpretation
Annual eligible attendances	72,000	Local activity report
Baseline median time to first assessment	84 minutes	Emergency-department dashboard
High-risk event prevalence	4.5%	Retrospective cohort
AI sensitivity at proposed threshold	0.88	External evidence; local verification required
AI specificity at proposed threshold	0.72	External evidence; local verification required
Incremental annual programme cost	EUR 310,000	Local lifecycle-cost estimate
Available additional rapid-assessment capacity	18 patients/day	Workforce and space assessment

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Map the present pathway and identify the decision window in which prioritisation could still change care.	Evidence and responsible owner to be recorded
2	Compare all realistic alternatives before assigning value to the AI option.	Evidence and responsible owner to be recorded
3	Represent the intervention bundle: data availability, interface, triage-nurse review, escalation, rapid-assessment capacity, monitoring, and fallback.	Evidence and responsible owner to be recorded
4	Estimate false-positive workload and false-negative consequences at the proposed threshold.	Evidence and responsible owner to be recorded
5	Test whether downstream capacity can absorb the additional patients identified as high risk.	Evidence and responsible owner to be recorded
6	Define a reversible pilot with clinical, operational, economic, equity, and trust indicators.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Expected high-risk alerts per day	Approximately 67	Includes true and false positive alerts
Additional patients requiring rapid review	Approximately 35/day	Exceeds currently available capacity without redesign
Primary value mechanism	Earlier review of high-risk patients	Dependent on response capacity
Main economic risk	Unusable alerts and displaced staff time	May increase rather than reduce delay
Preferred initial decision	Redesign and limited pilot	Not unrestricted deployment

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Proceed only with a redesigned, time-limited pilot that adds response capacity and compares the AI pathway with a staffing-and-streaming alternative. The model should remain advisory, with triage-nurse confirmation and a documented fallback.

- complete local calibration and silent-mode evaluation.
- demonstrate that rapid-assessment capacity can absorb expected alert volume.
- define subgroup and balancing measures, including delay among patients deprioritised by the system.
- cap total pilot expenditure and preserve the ability to stop without service disruption.
- review at 8 and 16 weeks using predefined proceed, redesign, restrict, or stop criteria.

Supplemental Resources

These supplemental resources translate the Chapter 1 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 1 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Intervention Specification	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Comparator and Opportunity-Cost Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Decision Conditions and Review Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Intervention Specification

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Comparator and Opportunity-Cost Record

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Map the present pathway and identify the decision window in which prioritisation could still change care. - Compare all realistic alternatives before assigning value to the AI option. - Represent the intervention bundle: data availability, interface, triage-nurse review, escalation, rapid-assessment capacity, monitoring, and fallback. - Estimate false-positive workload and false-negative consequences at the proposed threshold. - Test whether downstream capacity can absorb the additional patients identified as high risk. - Define a reversible pilot with clinical, operational, economic, equity, and trust indicators.

Decision Conditions and Review Record

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 1

Term	Working definition and decision relevance
Intervention bundle	Model, data, interface, people, workflow, capacity, governance, monitoring, and exit.
Comparator	The best realistic alternative use of the same resources.
Public value	Clinical, economic, organizational, equity, legitimacy, and stewardship effects considered together.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 1

Instrument	Status or type	Appropriate use and limitation
Health-system intervention	Analytical framing	Defines the complete unit of analysis.
Value chain	Causal map	Shows where capability may be converted into or lose value.
Economic evaluation	Method family	Compares costs and outcomes relative to a comparator.

Formula and Interpretation Note

The following relationship is used in Chapter 1 or its companion resources:

$$\text{Incremental value hypothesis: } \Delta V = f(\Delta O, \Delta C, \Delta W, \Delta R)$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE01 — Emergency-Department Triage Portfolio Decision Worked case; md; csv; json; xlsx; ods	Which intervention should the hospital test first when AI, staffing, and process redesign compete for the same limited resources?
AIHE-A01 — Decision and Use-Case Specification Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Define the service problem, full intervention, realistic comparators, outcomes, value mechanism, decision rule, and review cycle.
AIHE-A02 — Stakeholders, Decision Rights, and Participation Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Identify affected parties, decision rights, participation duties, evidence contributions, and escalation authority.
AIHE-RG07 — Common AI Adoption Failures and Corrective Actions Reference guide; docx; odt; xlsx; ods; csv; json; md	Identify recurring technology-led, economic, workflow, evidence, governance, procurement, and lifecycle failures and record preventive controls.
AIHE-EX01 — Project Intake and Stakeholder Register Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Capture the initiating problem, project scope, accountable owners, affected groups, and decision rights.
AIHE-W — AI Use-Case Canvas Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Define the problem, population, intended users, intervention bundle, comparator, outcomes, and initial risks.
AIHE-X — Decision Problem Template Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Specify population, intervention, comparator, outcomes, perspective, setting, time horizon, and decision rule.
AIHE-CASE00 — Common Case-Analysis Method Case-analysis method; md; docx; odt; xlsx; ods; csv; json	Provide a consistent problem-first, evidence, economic, governance, and lifecycle method for analysing worked and local cases.

Resource	Purpose
AIHE-AD — Fourteen-Week Course Schedule Optional teaching; docx; odt; xlsx; ods; csv; json; md	Provide an optional semester sequence for instructors using the book and companion resources.
AIHE-F — Suggested Twelve-Week Teaching Plan Optional teaching; docx; odt; xlsx; ods; csv; json; md	Provide a compressed optional teaching sequence separate from the institutional decision records.
AIHE-AG — Editable Figure and Graph Source Manifest Optional reference; docx; odt; xlsx; ods; csv; json; md	Record figure files, data, code, ownership, accessibility, licensing, version, and update status.
AIHE-LTX-01-REF — Reference Resources nan; TeX	Chapter 1 reference resource aligned with the current book structure.
AIHE-LTX-01-SUP — Supplemental Resources nan; TeX	Chapter 1 supplemental resource aligned with the current book structure.
AIHE-LTX-01-WOR — Emergency-Triage Intervention Portfolio nan; TeX	Chapter 1 worked-example resource aligned with the current book structure.

Chapter 2: Institutional Readiness and Digital Foundations

Chapter Orientation

Chapter 2 addresses the foundations that determine whether an AI-enabled service can operate reliably: data quality, interoperability, infrastructure, resilience, stewardship, and institutional capability. The resources distinguish general digital maturity from readiness for a specific use case and make shared foundation investments visible as portfolio assets.

Use this suite before local validation or procurement when source data, identity matching, terminology, interfaces, provenance, latency, security, or workflow ownership may constrain the proposed intervention. The output should show what is ready, what requires remediation, what can be tested safely, and which investments can be reused across future services.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Readmission Analytics Readiness and Data-Foundation Investment	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Use-Case Readiness Profile; Data and Interoperability Evidence Record; Foundation Investment and Portfolio Reuse	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	FHIR, OMOP, Digital maturity	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	2 core, 3 python analytics, 5 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Readmission Analytics Readiness and Data-Foundation Investment

Purpose and Status

A multi-site hospital group proposes a readmission-risk service, but discharge data, medication reconciliation, laboratory results, and prior admissions are distributed across several systems. The decision is whether to procure a model immediately or first strengthen reusable digital foundations.

Decision Problem and Comparators

Decision problem. Which sequence of investment offers the greatest institutional value: immediate procurement, a foundation-first programme, or a limited proof of concept restricted to one service?

Table: Comparators for the Chapter 2 worked example

Option	Comparator or strategy
1	Immediate vendor procurement
2	Foundation-first investment in identity, terminology, interfaces, provenance, and monitoring
3	Single-site proof of concept with manual data preparation
4	Conventional discharge-risk checklist and additional care-coordination capacity

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$G = \prod_{k=1}^{K} g_k, g_k \in \{0,1\}$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 2 worked example

Parameter	Base value	Source or interpretation
Hospitals in scope	4	Portfolio plan
Interfaces requiring remediation	11	Architecture inventory
Records with reliable medication reconciliation	68%	Data-quality profile
Cross-site patient-identity match rate	96.1%	Master-patient-index audit
Estimated foundation investment	EUR 740,000	Programme estimate
Use cases expected to reuse the foundations	7	Digital portfolio
Estimated procurement delay from foundation programme	9 months	Implementation plan

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Specify the data, timing, and workflow dependencies of the intended use.	Evidence and responsible owner to be recorded
2	Assess readiness by use case rather than by one maturity score.	Evidence and responsible owner to be recorded
3	Identify which weaknesses affect several proposed services and estimate their portfolio-level reuse value.	Evidence and responsible owner to be recorded
4	Compare immediate procurement with a foundation-first sequence and a restricted proof of concept.	Evidence and responsible owner to be recorded
5	Define acceptance criteria for identity, semantics, provenance, latency, resilience, and governance.	Evidence and responsible owner to be recorded
6	Record which standards are implementation tools and which local mappings and controls remain necessary.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Immediate procurement feasibility	Low	Critical data and identity dependencies unresolved
Foundation reuse potential	High	Seven planned services share the same capabilities
Short-term delay cost	Moderate	Potential benefit deferred for nine months
Risk of isolated procurement	High	Repeated mapping, weak provenance, and technical debt

Output	Result	Interpretation
Preferred decision	Foundation-first with restricted proof of concept	Allows learning without operational dependence

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Fund the shared digital-foundation programme and permit a restricted, non-clinically dependent proof of concept in one service. Procurement for routine use should wait until predefined data, interoperability, resilience, and governance thresholds are met.

- identity and duplicate-record controls meet the agreed threshold.
- medication and discharge data have documented provenance and completeness.
- interfaces meet decision-latency and recovery requirements.
- local FHIR profiles or equivalent exchange specifications and analytical transformations are version controlled.
- the investment case identifies portfolio reuse and avoids assigning all shared cost to one application.

Supplemental Resources

These supplemental resources translate the Chapter 2 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 2 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Use-Case Readiness Profile	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Data and Interoperability Evidence Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Foundation Investment and Portfolio Reuse	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Use-Case Readiness Profile

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Data and Interoperability Evidence Record

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Specify the data, timing, and workflow dependencies of the intended use. - Assess readiness by use case rather than by one maturity score. - Identify which weaknesses affect several proposed services and estimate their portfolio-level reuse value. - Compare immediate procurement with a foundation-first sequence and a restricted proof of concept. - Define acceptance criteria for identity, semantics, provenance, latency, resilience, and governance. - Record which standards are implementation tools and which local mappings and controls remain necessary.

Foundation Investment and Portfolio Reuse

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 2

Term	Working definition and decision relevance
FHIR	A standard for structured exchange of health information; it does not remove local profiling, mapping, or governance.
OMOP	A common data model for observational analysis; it is complementary to exchange standards.
Digital maturity	A profile of capabilities tied to a use case, not a certificate of universal readiness.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 2

Instrument	Status or type	Appropriate use and limitation
HIMSS maturity models	Maturity models	Road maps for capability development; not proof of readiness for every use.
FHIR	Interoperability standard	Supports exchange; requires local profiles and conformance.
OMOP CDM	Analytical data model	Supports observational analysis; does not replace source-system governance.
EHDS	Binding EU regulation with phased implementation	Shapes primary and secondary use, interoperability, access, and governance in scope.

Formula and Interpretation Note

The following relationship is used in Chapter 2 or its companion resources:

$$G = \prod_{k=1}^{K} g_k, g_k \in \{0, 1\}$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE02 — Readmission-Model Readiness and Data-Fitness Assessment Worked case; md; csv; json; xlsx; ods	Is the organization ready to validate and pilot a readmission-risk model, and which capability gaps must be remediated first?
AIHE-A03 — Institutional Readiness Assessment Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Assess strategic, data, infrastructure, workflow, workforce, quality, economic, governance, procurement, and sustainability readiness.
AIHE-A04 — Data Source, Provenance, and Fitness Record Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Document source and derived data assets, authority, definitions, transformations, quality, lineage, monitoring, and remediation.
AIHE-PY01 — Data quality, fitness-for-purpose, and mapping audit Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Profiles missingness, duplicates, implausible values, subgroup coverage, and terminology-mapping completeness before AI validation or economic analysis.
AIHE-PY03 — Missing-data sensitivity analysis Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Shows how complete-case analysis, simple imputation, and best/worst-case assumptions can change estimated outcomes and costs.
AIHE-PY33 — Patient or service segmentation using clustering	Uses standardized features and k-means clustering to identify groups with

Resource	Purpose
Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	different utilization, risk, cost, or workflow patterns.
AIHE-A — AI Readiness Assessment for Health Systems Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Assess strategic, data, infrastructure, workforce, governance, economic, ethical, security, and monitoring readiness at organizational and use-case levels.
AIHE-EX03 — Data Source Map and Access Record Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Document data sources, access authority, interoperability, quality, provenance, and reuse constraints.
AIHE-L — Health-System AI Readiness Maturity Assessment Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Rate health-system maturity by domain and identify the weakest critical capability and prerequisites before pilot or scale.
AIHE-U — SHARE-AI Secondary-Use Governance Worksheet Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Authorize secondary data use through purpose, harmonization, access, risk, public value, and audit controls.
AIHE-Y — Data Quality and Fitness-for-Purpose Assessment Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Assess completeness, plausibility, consistency, timeliness, representativeness, lineage, and actionability.
AIHE-LTX-02-REF — Reference Resources nan; TeX	Chapter 2 reference resource aligned with the current book structure.
AIHE-LTX-02-SUP — Supplemental Resources nan; TeX	Chapter 2 supplemental resource aligned with the current book structure.
AIHE-LTX-02-WOR — Readmission Analytics Readiness and Data-Foundation Investment nan; TeX	Chapter 2 worked-example resource aligned with the current book structure.

Chapter 3: Human–AI Decision Support and Managerial Action

Chapter Orientation

Chapter 3 focuses on the conversion of predictions, recommendations, alerts, and dashboard signals into accountable action. The resources specify decision windows, thresholds, users, human oversight, escalation, uncertainty, action latency, and the balancing measures required to detect overload or displaced burden.

Use this suite when designing clinical decision support, operational dashboards, command centres, or partially automated workflows. It helps prevent technically accurate information from being mistaken for decision value when users lack time, authority, context, or downstream capacity to act.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Hospital Command Dashboard and Patient Flow	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Human–AI Decision-Support Specification; Dashboard Signal-to-Action Record; Delegation, Oversight, and Action Latency	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Decision support, Delegation, Action latency	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	1 core, 4 python analytics, 1 supplemental specialist tool, 2 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Hospital Command Dashboard and Patient Flow

Purpose and Status

A tertiary hospital plans a command dashboard combining admissions, discharges, bed status, staffing, diagnostics, transport, and forecast demand. Leaders expect faster decisions, but current indicators have inconsistent definitions and no explicit action owners.

Decision Problem and Comparators

Decision problem. Should the hospital implement an AI-enhanced command dashboard, redesign the existing operational review process, or combine both through a staged signal-to-action intervention?

Table: Comparators for the Chapter 3 worked example

Option	Comparator or strategy
1	Current morning operational meeting and static reports
2	Standardized dashboard without predictive functions
3	AI-enhanced dashboard with demand forecasts and anomaly detection
4	Targeted process redesign for discharge, imaging, and transport bottlenecks

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$T_{action} = T_{delivery} + T_{review} + T_{response}$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 3 worked example

Parameter	Base value	Source or interpretation
Average daily admissions	312	Operations data
Mean bed occupancy	94%	Capacity dashboard
Median data latency	95 minutes	Interface audit
Indicators without named action owner	42%	Dashboard inventory
Forecast mean absolute percentage error	13%	Retrospective validation
Estimated annual dashboard cost	EUR 265,000	Lifecycle-cost estimate

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Define the operational decisions, decision windows, and accountable owners before selecting indicators.	Evidence and responsible owner to be recorded
2	Separate observed values, model estimates, forecasts, and recommendations.	Evidence and responsible owner to be recorded
3	Create an indicator dictionary with numerator, denominator, exclusions, latency, and limitations.	Evidence and responsible owner to be recorded
4	Map each signal to a feasible action and at least one balancing measure.	Evidence and responsible owner to be recorded
5	Test forecast value against a standardized non-predictive dashboard and targeted process redesign.	Evidence and responsible owner to be recorded
6	Measure action latency, downstream congestion, workload, and the consequences of recommendations that cannot be implemented.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Principal failure in current design	No action ownership	Information does not reliably change decisions
Forecast technical performance	Potentially useful	Requires local operational validation
Main value constraint	High occupancy and limited downstream capacity	Forecasting cannot create beds or staff
Preferred intervention	Signal-to-action redesign plus staged dashboard	Dashboard alone is insufficient
Primary monitoring outcome	Decision and action latency	Views and logins are secondary

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Implement a standardized operational dashboard first, with explicit action ownership and decision thresholds. Add forecasting functions only after the data-latency and response-pathway requirements have been met and tested in simulation and a controlled pilot.

- all priority indicators have definitions, owners, decision windows, and balancing measures.
- data latency remains within the operational window.
- forecast uncertainty is displayed and observed values are distinguishable from predictions.
- the pilot measures action and outcome changes rather than views or user satisfaction alone.
- a governance process reviews gaming, surveillance concerns, and unequal workload effects.

Supplemental Resources

These supplemental resources translate the Chapter 3 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 3 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Human–AI Decision-Support Specification	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Dashboard Signal-to-Action Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Delegation, Oversight, and Action Latency	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Human–AI Decision-Support Specification

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Dashboard Signal-to-Action Record

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define the operational decisions, decision windows, and accountable owners before selecting indicators. - Separate observed values, model estimates, forecasts, and recommendations. - Create an indicator dictionary with numerator, denominator, exclusions, latency, and limitations. - Map each signal to a feasible action and at least one balancing measure. - Test forecast value against a standardized non-predictive dashboard and targeted process redesign. - Measure action latency, downstream congestion, workload, and the consequences of recommendations that cannot be implemented.

Delegation, Oversight, and Action Latency

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 3

Term	Working definition and decision relevance
Decision support	Information intended to influence a defined decision and action within a meaningful time window.
Delegation	The allocation of tasks and authority between people and technology.
Action latency	The elapsed time from signal generation to feasible response.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 3

Instrument	Status or type	Appropriate use and limitation
DECIDE-AI	Reporting guideline	Supports transparent reporting of early live clinical evaluation.
Decision-curve analysis	Analytical method	Examines threshold-specific net benefit.
Safety case	Assurance argument	Links hazards, controls, and evidence for a defined context.

Formula and Interpretation Note

The following relationship is used in Chapter 3 or its companion resources:

$$T_{action} = T_{delivery} + T_{review} + T_{response}$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE03 — Deterioration Alert Under Constrained Review Capacity Worked case; md; csv; json; xlsx; ods	Which alert threshold produces an operationally safe balance between missed deterioration and review burden?
AIHE-CASE03B — Hospital Command Dashboard and Patient Flow Worked case; md; csv; json; xlsx; ods	Evaluate whether and under what conditions a command dashboard should be scaled as a permanent patient-flow management intervention.
AIHE-C04 — Human Oversight and Accountability Matrix Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Assign accountability, responsibility, consultation, information, escalation, pause, update, restart, and retirement authority.
AIHE-PY06 — Decision-curve analysis Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Compares prediction-model net benefit with treat-all and treat-none strategies across clinically relevant thresholds.
AIHE-PY21 — Discrete-event simulation of hospital flow Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Simulates arrivals, queues, service times, capacity, prioritization, and delays to assess how AI-enabled triage or scheduling affects operations.
AIHE-PY34 — Time-series forecasting for demand, capacity, and budget planning Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Fits simple trend and exponential-smoothing forecasts to operational or financial time series and compares forecast accuracy.
AIHE-PY35 — Constrained resource-allocation optimization Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Selects an allocation across candidate AI or service options to maximize expected value subject to budget, workforce, capacity, and minimum-service constraints.
AIHE-O — HOAM Accountability Workbook Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Assign accountable and responsible owners, consultation, information, escalation, pause, update, restart, and retirement authority.
AIHE-LTX-03-REF — Reference Resources nan; TeX	Chapter 3 reference resource aligned with the current book structure.

Resource	Purpose
AIHE-LTX-03-SUP — Supplemental Resources nan; TeX	Chapter 3 supplemental resource aligned with the current book structure.
AIHE-LTX-03-WOR — Hospital Command Dashboard and Patient Flow nan; TeX	Chapter 3 worked-example resource aligned with the current book structure.

Part II: Organizational Performance, Workforce, and Quality

This Part examines whether proposed capability can be converted into productive work, quality, accreditation evidence, patient safety, and organizational learning.

Chapter 4: Productivity, Workforce Redesign, and Implementation

Chapter Orientation

Chapter 4 connects productivity claims with workforce redesign and implementation. The resources separate gross time saved from review and correction, usable capacity, capacity actually redeployed, quality-adjusted outcomes, and the implementation work required to sustain benefit.

Use this suite for documentation tools, scheduling systems, administrative automation, and other proposals built around efficiency. It supports task decomposition, workload-transfer analysis, implementation strategy selection, training, adaptation, sustainment, and a defensible benefit-realization plan.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Ambient Documentation and Workforce Redesign	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Productivity-Realization Chain; Determinant-to-Strategy Matrix; Implementation and Sustainment Record	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Productivity, Implementation determinant, Sustainment	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	1 core, 3 python analytics, 1 reference guide, 2 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Ambient Documentation and Workforce Redesign

Purpose and Status

An ambulatory network is considering an ambient documentation system to reduce after-hours work. The vendor reports substantial drafting-time savings, while clinicians are concerned about correction burden, privacy, coding changes, and pressure to convert every saved minute into additional appointments.

Decision Problem and Comparators

Decision problem. Does the intervention create net workforce and organizational value, and what workflow, governance, and implementation conditions are required before scale?

Table: Comparators for the Chapter 4 worked example

Option	Comparator or strategy
1	Current documentation workflow
2	Template and workflow redesign without generative technology
3	Additional transcription or administrative support
4	Ambient documentation with opt-in human review

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$T_{net} = T_{gross} - T_{review} - T_{correction} - T_{coordination}$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 4 worked example

Parameter	Base value	Source or interpretation
Clinicians in pilot	60	Pilot plan
Baseline documentation time per visit	11.5 minutes	Time study
Vendor-reported gross saving	5.0 minutes/visit	Vendor evidence
Observed review and correction time	2.4 minutes/visit	Local usability test
Monthly subscription per clinician	EUR 420	Proposed contract
Average visits per clinician per month	310	Scheduling system
Clinically consequential correction rate	1.8%	Pilot quality audit

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Map the full documentation workflow and distinguish drafting assistance from coding or downstream recommendations.	Evidence and responsible owner to be recorded
2	Measure gross time, review, correction, escalation, and coordination by professional role.	Evidence and responsible owner to be recorded
3	Assess whether net time can be pooled or scheduled as usable capacity.	Evidence and responsible owner to be recorded
4	Evaluate quality, privacy, consent, patient interaction, and after-hours work alongside throughput.	Evidence and responsible owner to be recorded
5	Use CFIR, NASSS-CAT, or another selected implementation resource for a defined purpose rather than as a generic checklist.	Evidence and responsible owner to be recorded
6	Define benefit owners, renewal thresholds, and controls for model, prompt, and template updates.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Gross time saved	5.0 minutes/visit	Vendor claim
Net time saved after review	2.6 minutes/visit	Local pilot estimate
Potential monthly net time per clinician	13.4 hours	Before accounting for downtime and meetings
Cash-releasing saving	Not established	No payroll reduction demonstrated
Main realized value candidate	Reduced after-hours work and improved patient interaction	Requires longitudinal evidence

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Continue a limited opt-in pilot and do not treat gross time saving as a cash benefit. Scale only if net time benefit, documentation quality, patient acceptability, privacy, and workforce experience remain favourable and no adverse coding or downstream-utilization signal emerges.

- documentation assistance remains separate from coding and order automation.
- clinically consequential content requires meaningful professional confirmation.
- specialty-specific correction patterns and subgroup effects are monitored.
- released time has a documented and acceptable use.
- renewal is linked to sustained net benefit after the novelty period.

Supplemental Resources

These supplemental resources translate the Chapter 4 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 4 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Productivity-Realization Chain	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Determinant-to-Strategy Matrix	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Implementation and Sustainment Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Productivity-Realization Chain

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Determinant-to-Strategy Matrix

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Map the full documentation workflow and distinguish drafting assistance from coding or downstream recommendations. - Measure gross time, review, correction, escalation, and coordination by professional role. - Assess whether net time can be pooled or scheduled as usable capacity. - Evaluate quality, privacy, consent, patient interaction, and after-hours work alongside throughput. - Use CFIR, NASSS-CAT, or another selected implementation resource for a defined purpose rather than as a generic checklist. - Define benefit owners, renewal thresholds, and controls for model, prompt, and template updates.

Implementation and Sustainment Record

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 4

Term	Working definition and decision relevance
Productivity	Output or value generated relative to resources, adjusted for quality and transferred work.
Implementation determinant	A factor that impedes or enables implementation in a defined context.
Sustainment	Continued appropriate use and benefit after initial implementation support declines.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 4

Instrument	Status or type	Appropriate use and limitation
MRC guidance	Methodological guidance	Supports development and evaluation of complex interventions.
CFIR	Determinant framework	Identifies barriers and facilitators.
NASSS-CAT	Complexity assessment tool	Examines nonadoption, abandonment, scale, spread, and sustainability.
RE-AIM/PRISM	Implementation-outcome and context frameworks	Support real-world implementation and maintenance evaluation.

Formula and Interpretation Note

The following relationship is used in Chapter 4 or its companion resources:

$$T_{net} = T_{gross} - T_{review} - T_{correction} - T_{coordination}$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE04 — Ambient Documentation Assistant: Productivity and Workforce Redesign Worked case; md; csv; json; xlsx; ods	How much of the claimed documentation time saving becomes usable capacity after review, correction, adoption, and implementation costs?
AIHE-B03 — Benefit Realization and Capacity Conversion Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Translate claimed benefits into measurable mechanisms, owners, baselines, conversion conditions, and realized results.
AIHE-RG03 — Implementation Framework Selection Guide Reference guide; docx; odt; xlsx; ods; csv; json; md	Select CFIR, NASSS-CAT, RE-AIM/PRISM, TAM/UTAUT, and mixed-method approaches according to the implementation question rather than name recognition.
AIHE-PY22 — System-dynamics model of adoption, capacity, and value Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Explores feedback among training, trust, adoption, workforce capacity, workload, benefit realization, and support burden over time.
AIHE-PY23 — Agent-based simulation of clinician adoption Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Models heterogeneous clinicians whose adoption changes through training, peer influence, perceived usefulness, workload, and adverse incidents.
AIHE-PY28 — Workflow time, capacity, productivity, and benefit realization Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Translates gross task-time savings into net time, usable capacity, realized output, quality-adjusted productivity, and economic value.
AIHE-EX07 — Benefit Realization and Capacity Conversion Log Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd;	Assign each benefit to a mechanism, measure, owner, baseline, timing, accounting treatment, and verification source.

Resource	Purpose
svg; png	
AIHE-Z — ROI and Workflow Value Worksheet Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Distinguish gross time saved, review burden, capacity released, capacity used, and realized value.
AIHE-LTX-04-REF — Reference Resources nan; TeX	Chapter 4 reference resource aligned with the current book structure.
AIHE-LTX-04-SUP — Supplemental Resources nan; TeX	Chapter 4 supplemental resource aligned with the current book structure.
AIHE-LTX-04-WOR — Ambient Documentation and Workforce Redesign nan; TeX	Chapter 4 worked-example resource aligned with the current book structure.

Chapter 5: Quality, Accreditation, and Organizational Learning

Chapter Orientation

Chapter 5 places AI-enabled services within quality management, accreditation, patient safety, corrective action, audit, and organizational learning. The resources link technical and clinical indicators with workflow, equity, patient experience, economic value, incident investigation, and verified corrective action.

Use this suite after a pilot or deployment, after a model or workflow update, or when monitoring identifies an unexpected signal. It helps distinguish statistical variation from unacceptable performance and connects investigation, containment, corrective action, revalidation, communication, and closure.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Quality Signal After a Model and Workflow Update	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Quality Monitoring Plan; Incident and CAPA Record; Accreditation and Organizational-Learning Record	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Quality assurance, Quality control, Organizational learning	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	2 core, 1 python analytics, 1 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Quality Signal After a Model and Workflow Update

Purpose and Status

A deterioration-alert service has been used for eighteen months. Following an electronic health-record upgrade and a model update, the false-alert rate rises and nurses report more workarounds, while aggregate discrimination appears unchanged.

Decision Problem and Comparators

Decision problem. Should the service continue unchanged, be restricted, undergo corrective action and revalidation, or be suspended pending investigation?

Table: Comparators for the Chapter 5 worked example

Option	Comparator or strategy
1	Continue current operation
2	Revert to the previous model and interface version
3	Restrict use to selected units
4	Suspend and return to the conventional escalation pathway

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$R_t = \frac{\text{events meeting the quality definition in period } t}{\text{eligible opportunities in period } t}$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 5 worked example

Parameter	Base value	Source or interpretation
Baseline false-alert rate	14%	Quality dashboard
Post-update false-alert rate	23%	Four-week monitoring window
Aggregate AUROC before/after	0.82 / 0.81	Monitoring report
Median response time before/after	12 / 19 minutes	Workflow log
Units reporting workarounds	5 of 8	Safety walk-rounds
Serious incidents	0	Incident register
Near-miss reports	7	Patient-safety system

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Reconstruct the change timeline, data sources, model, software, interface, threshold, and workflow versions.	Evidence and responsible owner to be recorded
2	Distinguish statistical stability from clinical and managerial acceptability.	Evidence and responsible owner to be recorded
3	Investigate source-data, interface, threshold, user, staffing, and downstream-capacity explanations before selecting corrective action.	Evidence and responsible owner to be recorded
4	Use the existing quality and patient-safety system to document containment, root causes, corrective and preventive action, and effectiveness checks.	Evidence and responsible owner to be recorded
5	Assess whether the change is material and which validation evidence must be repeated.	Evidence and responsible owner to be recorded
6	Communicate the decision, residual uncertainty, and conditions for return to unrestricted use.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Technical discrimination	Broadly stable	Does not rule out workflow or calibration failure
Operational performance	Deteriorated	Response delay and alert burden increased
Likely cause	Combined interface and threshold effects	Requires confirmation
Immediate action	Restrict and revert where feasible	Containment while investigation proceeds
Quality decision	CAPA and targeted revalidation	Not automatic retraining

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Restrict use, restore the previous validated configuration where safe, and complete a formal incident and corrective-action process. Unrestricted use should resume only after the source of the signal is understood and the effectiveness of corrective action is verified.

- versioned reconstruction of data, software, model, interface, and workflow.
- unit-level and subgroup analysis.
- documented CAPA with named owners and completion dates.
- verification that response time and alert burden return within acceptable limits.
- executive authority to suspend or retire the intervention if performance remains unacceptable.

Supplemental Resources

These supplemental resources translate the Chapter 5 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 5 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Quality Monitoring Plan	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Incident and CAPA Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Accreditation and Organizational-Learning Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Quality Monitoring Plan

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Incident and CAPA Record

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Reconstruct the change timeline, data sources, model, software, interface, threshold, and workflow versions. - Distinguish statistical stability from clinical and managerial acceptability. - Investigate source-data, interface, threshold, user, staffing, and downstream-capacity explanations before selecting corrective action. - Use the existing quality and patient-safety system to document containment, root causes, corrective and preventive action, and effectiveness checks. - Assess whether the change is material and which validation evidence must be repeated. - Communicate the decision, residual uncertainty, and conditions for return to unrestricted use.

Accreditation and Organizational-Learning Record

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 5

Term	Working definition and decision relevance
Quality assurance	Evidence supporting confidence that requirements are being met.
Quality control	Operational detection of variation or failure against statistical and acceptability limits.
Organizational learning	Conversion of monitoring and experience into revised knowledge, practice, and governance.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 5

Instrument	Status or type	Appropriate use and limitation
ISO 7101	Voluntary healthcare management-system standard	Supports quality-management requirements.
ISO/IEC 42001	Voluntary AI management-system standard	Supports organization-wide AI management responsibilities.
SPC	Statistical method	Supports detection of process variation; not a clinical acceptability rule.
CAPA	Quality process	Links investigation to corrective and preventive action and verification.

Formula and Interpretation Note

The following relationship is used in Chapter 5 or its companion resources:

$$R_t = \frac{\text{events meeting the quality definition in period } t}{\text{eligible opportunities in period } t}$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE05 — AI Quality-Control, Accreditation, and Corrective Action Worked case; md; csv; json; xlsx; ods	When should a change in performance trigger investigation, restriction, and corrective action within the quality-management system?
AIHE-C01 — AI Quality-Control and Accreditation Evidence Plan Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Integrate AI-enabled services into quality control, accreditation evidence, monitoring, thresholds, action, and closure.
AIHE-C02 — Corrective and Preventive Action Record Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Document nonconformity, containment, causal analysis, corrective and preventive action, effectiveness, residual risk, and closure.
AIHE-PY02 — Descriptive statistics and visual comparisons Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Summarizes numerical and categorical variables, compares groups, and creates baseline visual diagnostics before modeling.
AIHE-I — Health-System AI Indicator Library Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Select outcome, process, balancing, equity, workforce, economic, governance, environmental, and lifecycle indicators with definitions, owners, sources, thresholds, and actions.
AIHE-LTX-05-REF — Reference Resources nan; TeX	Chapter 5 reference resource aligned with the current book structure.
AIHE-LTX-05-SUP — Supplemental Resources nan; TeX	Chapter 5 supplemental resource aligned with the current book structure.

Resource	Purpose
AIHE-LTX-05-WOR — Quality Signal After a Model and Workflow Update nan; TeX	Chapter 5 worked-example resource aligned with the current book structure.

Part III: Economic Evaluation, Decision Modelling, and Procurement

This Part develops lifecycle costing, economic evaluation, modelling under uncertainty, procurement, contracting, and vendor governance.

Chapter 6: Lifecycle Cost, Pricing, and Benefit Realization

Chapter Orientation

Chapter 6 develops the full lifecycle cost and financial case. The resources identify costs that sit outside the licence price, model volume and pricing exposure, distinguish cash savings from avoided cost and released capacity, and assign responsibility for converting projected benefit into observed value.

Use this suite during business-case development, pricing negotiation, pilot budgeting, renewal, or exit. The records should preserve the perspective, price year, time horizon, cost boundary, allocation rules, volume scenarios, benefit mechanism, budget owner, uncertainty, and residual value.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Clinical Documentation Platform: Lifecycle Cost and Pricing	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Lifecycle Cost Log; Pricing and Volume Scenarios; Benefit-Realization Register	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Lifecycle cost, Pricing exposure, Benefit realization	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	2 core, 2 python analytics, 3 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Clinical Documentation Platform: Lifecycle Cost and Pricing

Purpose and Status

A hospital group is comparing three pricing proposals for a documentation platform: a fixed enterprise licence, a per-clinician subscription, and a hybrid base fee plus per-encounter charge. Each proposal excludes some implementation, monitoring, and exit activities.

Decision Problem and Comparators

Decision problem. Which pricing and contracting structure produces the most predictable and defensible lifecycle cost, and under what conditions could projected benefits be realized?

Table: Comparators for the Chapter 6 worked example

Option	Comparator or strategy
1	Fixed enterprise licence
2	Per-clinician subscription
3	Hybrid base plus encounter-based fee
4	Do not procure; invest in workflow and template redesign

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$ROI = \frac{PV(B) - PV(C)}{PV(C)} \times 100\%$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 6 worked example

Parameter	Base value	Source or interpretation
Clinicians	420	Workforce plan
Annual eligible encounters	520,000	Activity forecast
Enterprise base fee	EUR 1.15 million/year	Vendor A
Per-clinician fee	EUR 390/month	Vendor B
Hybrid base fee	EUR 540,000/year	Vendor C
Hybrid variable fee	EUR 1.15/encounter	Vendor C
Initial integration and governance	EUR 760,000	Local estimate
Annual monitoring and support	EUR 310,000	Local estimate
Expected useful decision horizon	5 years	Business case

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Define perspective, price year, time horizon, comparator, and full cost boundary.	Evidence and responsible owner to be recorded
2	Classify costs as fixed, step-fixed, variable, contingent, shared, or exit related.	Evidence and responsible owner to be recorded
3	Model low, expected, and high activity for each pricing structure.	Evidence and responsible owner to be recorded
4	Discount costs and credible monetised benefits; preserve non-monetised value separately.	Evidence and responsible owner to be recorded
5	Create a benefit-realization register that distinguishes cash savings, avoided costs, usable capacity, quality, workforce, and public value.	Evidence and responsible owner to be recorded
6	Test price, uptake, review burden, implementation delay, support, and exit thresholds.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Most predictable annual spend	Enterprise licence	May overpay at low uptake
Lowest expected five-year PV cost	Hybrid proposal	Dependent on activity cap and included support
Dominant uncertainty	Net usable capacity per encounter	Not vendor price alone
Cash-releasing saving	Not established	Released time is not yet linked to budget reduction
Preferred decision	Negotiate hybrid with cap and conditional renewal	Preserves predictability and proportionality

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Negotiate a capped hybrid structure with explicit integration, monitoring, updates, data access, and exit support. Treat productivity benefit as conditional until local evidence shows how net time is converted into usable capacity or another measurable outcome.

- all pricing units are clinically and financially understandable.
- base and variable components do not double charge essential services.
- low, expected, and high activity scenarios remain affordable.
- benefit owners and verification sources are named.
- renewal depends on observed value, current evidence, and acceptable exit options.

Supplemental Resources

These supplemental resources translate the Chapter 6 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 6 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Lifecycle Cost Log	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Pricing and Volume Scenarios	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Benefit-Realization Register	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Lifecycle Cost Log

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Pricing and Volume Scenarios

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define perspective, price year, time horizon, comparator, and full cost boundary. - Classify costs as fixed, step-fixed, variable, contingent, shared, or exit related. - Model low, expected, and high activity for each pricing structure. - Discount costs and credible monetised benefits; preserve non-monetised value separately. - Create a benefit-realization register that distinguishes cash savings, avoided costs, usable capacity, quality, workforce, and public value. - Test price, uptake, review burden, implementation delay, support, and exit thresholds.

Benefit-Realization Register

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 6

Term	Working definition and decision relevance
Lifecycle cost	Incremental resources from preparation through operation, change, transition, and exit.
Pricing exposure	How contractual units and activity translate into financial risk.
Benefit realization	Evidence that a projected mechanism produced an observable and usable result.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 6

Instrument	Status or type	Appropriate use and limitation
TCAIO	Author-developed costing framework	Organizes full lifecycle cost from acquisition through exit.
CHEERS-AI	Reporting guideline	Supports transparent reporting of AI economic evaluations.
TDABC	Costing method	Estimates personnel and activity costs from time and capacity cost rates.

Formula and Interpretation Note

The following relationship is used in Chapter 6 or its companion resources:

$$ROI = \frac{PV(B) - PV(C)}{PV(C)} \times 100\%$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE06 — Five-Year Lifecycle Cost, Pricing, ROI, and Benefit Realization Worked case; md; csv; json; xlsx; ods	Does the proposed AI service create value after all lifecycle costs, pricing exposure, and benefit-conversion conditions are included?
AIHE-B01 — Economic Evaluation Specification Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Define the economic decision, perspective, comparator, horizon, outcomes, costs, methods, uncertainty, safeguards, and validation.
AIHE-B02 — Lifecycle Cost and Pricing Record Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Record acquisition through exit costs, resource use, pricing exposure, opportunity cost, residual value, and evidence.
AIHE-PY10 — Total Cost of AI Ownership (TCAIO) lifecycle costing Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Calculates one-time and recurring AI costs, discounted present value, cost per use, and optimistic/base/conservative scenarios across the full lifecycle.
AIHE-PY11 — ROI, net benefit, and benefit-realization analysis Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Calculates gross and risk-adjusted benefits, total cost, net benefit, ROI, payback period, and the effect of realization probability and attribution assumptions.
AIHE-B — Economic Evaluation Specification Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Specify the decision, perspective, comparator, costs, outcomes, horizon, uncertainty, distribution, and reporting plan.
AIHE-EX17 — AI Pricing and Contract Scenario Record Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Model base and variable fees in clinically legible units, cap exposure, price support explicitly, and link renewal to realized value.
AIHE-M — TCAIO Costing Workbook Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Estimate incremental lifecycle costs from acquisition through operation, monitoring, material change, and exit.
AIHE-LTX-06-REF — Reference Resources nan; TeX	Chapter 6 reference resource aligned with the current book structure.
AIHE-LTX-06-SUP — Supplemental Resources nan; TeX	Chapter 6 supplemental resource aligned with the current book structure.

Resource	Purpose
AIHE-LTX-06-WOR — Clinical Documentation Platform: Lifecycle Cost and Pricing nan; TeX	Chapter 6 worked-example resource aligned with the current book structure.

Chapter 7: Economic Evaluation, Affordability, and HTA

Chapter Orientation

Chapter 7 separates value for money, budget impact, affordability, health technology assessment, reimbursement, and local adoption. The resources support a transparent economic-evaluation protocol and ensure that cost-effectiveness does not obscure cash-flow constraints, implementation capacity, distributional effects, or unresolved safeguards.

Use this suite when an institution must decide whether to adopt, pilot, defer, redesign, restrict, or reject an intervention. The outputs should show incremental costs and outcomes, annual budget consequences, evidence maturity, uncertainty, subgroup effects, payment context, and local prerequisites.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Remote Monitoring: Economic Evaluation, Affordability, and HTA	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Economic Evaluation Protocol; Budget-Impact Record; HTA and Local-Adoption Decision Record	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Cost-effectiveness, Affordability, HTA	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	1 core, 3 python analytics, 2 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Remote Monitoring: Economic Evaluation, Affordability, and HTA

Purpose and Status

A public payer and hospital network are assessing an AI-supported remote-monitoring programme for people with heart failure. The programme may reduce emergency visits and admissions, but it requires devices, connectivity, monitoring staff, escalation capacity, and recurring platform fees.

Decision Problem and Comparators

Decision problem. Is the programme cost-effective, affordable, and suitable for local adoption, and which evidence gaps justify a staged decision?

Table: Comparators for the Chapter 7 worked example

Option	Comparator or strategy
1	Usual scheduled follow-up
2	Non-AI remote monitoring with threshold rules
3	AI-supported remote monitoring with nurse review
4	Enhanced community nursing without remote monitoring

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$NMB = \lambda \Delta E - \Delta C$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 7 worked example

Parameter	Base value	Source or interpretation
Eligible population	8,000	Commissioning estimate
Year-1 uptake	35%	Implementation scenario
Incremental cost per participant	EUR 1,180/year	Programme costing
Expected admissions avoided	0.10/person-year	Evidence synthesis
QALY gain	0.028/person-year	Economic model
Cost per admission	EUR 5,600	Payer data
Implementation cost	EUR 1.4 million	Network estimate
Decision threshold	EUR 35,000/QALY	Illustrative local threshold

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Specify decision-maker, perspective, comparator, population, horizon, and intended scale.	Evidence and responsible owner to be recorded
2	Estimate incremental costs, effects, QALYs, and net benefit.	Evidence and responsible owner to be recorded
3	Present annual and cumulative budget impact under uptake scenarios.	Evidence and responsible owner to be recorded
4	Separate economic cost from cash flow and productive capacity from cash-releasing savings.	Evidence and responsible owner to be recorded
5	Assess subgroup access, connectivity, digital support, and distribution of opportunity cost.	Evidence and responsible owner to be recorded
6	Identify whether remaining uncertainty should lead to adoption, a restricted pilot, price negotiation, or further research.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Incremental programme cost before offsets	EUR 9.44 million/year at full scale	Excludes initial implementation
Expected admission offset	EUR 4.48 million/year at full scale	Dependent on causal effectiveness
Illustrative ICER	Approximately EUR 22,900/QALY	Favourable at stated threshold
Year-1 budget impact	Positive and substantial	May remain unaffordable despite cost-effectiveness
Preferred decision	Phased adoption with negotiated price and access support	HTA and affordability separated

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Support phased adoption where the comparative clinical evidence is credible, but link scale to annual affordability, observed service capacity, subgroup access, and updated budget impact. The payer and provider should align payment and implementation responsibilities.

- the exact intervention bundle and comparator are defined.
- annual and cumulative budget impact remain within an agreed envelope.
- non-digital access and connectivity support are funded.
- local causal and implementation evidence is collected.
- price, uptake, and service-capacity thresholds are included in the decision record.

Supplemental Resources

These supplemental resources translate the Chapter 7 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 7 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Economic Evaluation Protocol	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Budget-Impact Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
HTA and Local-Adoption Decision Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Economic Evaluation Protocol

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Budget-Impact Record

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Specify decision-maker, perspective, comparator, population, horizon, and intended scale. - Estimate incremental costs, effects, QALYs, and net benefit. - Present annual and cumulative budget impact under uptake scenarios. - Separate economic cost from cash flow and productive capacity from cash-releasing savings. - Assess subgroup access, connectivity, digital support, and distribution of opportunity cost. - Identify whether remaining uncertainty should lead to adoption, a restricted pilot, price negotiation, or further research.

HTA and Local-Adoption Decision Record

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 7

Term	Working definition and decision relevance
Cost-effectiveness	Comparison of incremental outcomes and costs relative to an alternative.
Affordability	Whether the responsible budget and implementation capacity can support the intervention.
HTA	Multidisciplinary appraisal supporting policy, reimbursement, and adoption decisions.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 7

Instrument	Status or type	Appropriate use and limitation
CHEERS 2022	Reporting guideline	Supports transparent reporting of health-economic evaluations.
CHEERS-AI	AI-specific reporting extension	Adds AI intervention, implementation, updating, and uncertainty considerations.
NICE reference case	Jurisdiction-specific methods guidance	Supports consistent appraisal in England; not universal.
EU HTA Regulation	Binding EU regulation in scope	Supports joint clinical assessment while national decisions remain.

Formula and Interpretation Note

The following relationship is used in Chapter 7 or its companion resources:

$$NMB = \lambda \Delta E - \Delta C$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE07 — Remote Monitoring: Cost-Effectiveness, Budget Impact, and HTA Worked case; md; csv; json; xlsx; ods	Can an intervention be cost-effective yet remain unaffordable or unsuitable for local adoption?
AIHE-B04 — Cost-Effectiveness, Budget Impact, and HTA Record Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Summarize cost-effectiveness, affordability, distribution, HTA, reimbursement, adoption conditions, uncertainty, and recommendation.
AIHE-PY12 — Cost-effectiveness, ICER, and net monetary benefit Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Compares an AI-enabled pathway with a comparator using incremental cost, incremental effect, ICER, net monetary benefit, and dominance rules.
AIHE-PY13 — Multiyear budget-impact analysis Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Estimates annual and cumulative financial consequences for a specific budget holder under low, base, and high adoption or effectiveness scenarios.
AIHE-PY16 — Distributional cost-effectiveness and equity-weighted analysis Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Compares costs, effects, net monetary benefit, access, and equity-weighted value across population groups.
AIHE-AA — Cost-Effectiveness and Budget-Impact Tool Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Compare incremental costs and outcomes, net benefit, affordability, uptake, cash flow, and sensitivity scenarios.
AIHE-EX18 — HTA, Reimbursement, and Adoption Record Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Distinguish regulatory authorization, HTA, reimbursement or payment, procurement, and local adoption; record the authority, perspective, evidence, gaps, and continuing evidence conditions for each layer.
AIHE-LTX-07-REF — Reference Resources nan; TeX	Chapter 7 reference resource aligned with the current book structure.
AIHE-LTX-07-SUP — Supplemental Resources nan; TeX	Chapter 7 supplemental resource aligned with the current book structure.

Resource	Purpose
AIHE-LTX-07-WOR — Remote Monitoring: Economic Evaluation, Affordability, and HTA nan; TeX	Chapter 7 worked-example resource aligned with the current book structure.

Chapter 8: Decision Modelling and Simulation Under Uncertainty

Chapter Orientation

Chapter 8 explains how the structure of the decision determines the modelling method. The resources cover decision trees, state-transition models, microsimulation, discrete-event simulation, queueing, hybrid models, scenario analysis, probabilistic sensitivity analysis, and value of information.

The detailed imaging-prioritisation case is retained as the principal worked example because it shows how queue effects, classification errors, time to diagnosis, capacity, lifetime outcomes, subgroup consequences, and research priorities can be linked without treating image-level performance as the whole decision.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Imaging Prioritisation	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Model Specification; Verification, Validation, and Reproducibility; Uncertainty and Value-of-Information Record	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Decision model, Simulation, Value of information	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	1 core, 5 python analytics, 1 reference guide, 5 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Imaging Prioritisation

Purpose and Status

This worked example shows how a health system can combine queue modelling, diagnostic consequences, economic evaluation, subgroup analysis, and decision uncertainty when considering an artificial intelligence (AI) system that prioritises imaging studies for human interpretation. The intervention changes the order in which studies are read; it does not replace professional review or issue a final diagnosis.

All values are synthetic and are included solely to demonstrate the analytical sequence. They are not clinical estimates, price benchmarks, or recommendations for a particular imaging service. Local use would require current clinical evidence, local workflow and capacity data, subgroup validation, approved cost sources, and governance review. The recent United Kingdom breast-screening evaluation cited in Chapter 8 illustrates the value of connecting operational workflow with longer-term health and economic consequences, but the numerical values below are not taken from that study (1).

Field	Specification
Resource identifier	AIHE-CASE08
Aligned chapter	Chapter 8: Decision Modelling and Simulation Under Uncertainty
Decision setting	Annual cohort of screening-imaging examinations
Perspective	Hospital and publicly funded health-system perspective
Price year	2026 pounds sterling (GBP)
Analytical horizon	One year of service operation, with lifetime consequences assigned to materially delayed cancer diagnoses
Comparator	Current first-in, first-out worklist with manual escalation when staff identify urgency
Intervention	AI-supported prioritisation followed by unchanged human interpretation and diagnostic responsibility
Status	Synthetic worked example; not a validated institutional model

Decision Problem and Comparator

A regional imaging service processes approximately 25,000 screening studies each year. The organisation is considering an AI prioritisation system that assigns a priority flag when an examination appears more likely to require rapid review. Flagged studies move to the front of the human-reading worklist. All studies are still interpreted by qualified readers, and the intervention does not alter the number of readers, scanners, diagnostic appointments, or treatment slots.

The decision is whether to retain the current worklist, conduct a controlled pilot, or adopt the prioritisation system at scale. The analysis must determine whether earlier review of time-sensitive cases produces sufficient health and resource value to justify licensing, integration, monitoring, training, and the opportunity cost imposed on non-prioritised studies.

Decision element	Worked specification
Population	Adults receiving screening imaging during one service year
Service problem	Clinically important studies may remain in the routine queue long enough to reduce the benefit of early diagnosis
Comparator	First-in, first-out worklist with existing manual escalation
Intervention	AI priority flag, priority queue, human interpretation, audit, monitoring, and rollback capability
Excluded uses	Autonomous diagnosis, removal of human review, automated treatment decisions, or use outside the validated imaging pathway
Primary outcome	Quality-adjusted life years (QALYs) gained by reducing materially delayed diagnoses
Operational outcomes	Time to report, priority volume, routine-worklist delay, capacity use, and review burden
Economic outcomes	Incremental cost, incremental QALYs, incremental cost-effectiveness ratio, net monetary benefit, and budget exposure
Equity question	Whether performance and reporting time differ for groups with more complex studies or poorer access to specialist services
Decision options	Do not adopt; redesign; conduct a controlled pilot; adopt with conditions; scale; restrict; or discontinue

Model Structure

The worked example uses a simplified hybrid structure. Each component answers a different part of the decision problem: - A *discrete-event simulation* (DES) represents arrivals, daily reading capacity, temporary capacity loss, queue order, and time to report. DES is appropriate because prioritisation changes the ordering of events and the use of constrained resources (2). - A short *decision tree* classifies examinations as true positive, false negative, false positive, or true negative relative to the priority flag. These categories determine which studies receive earlier review and which studies consume priority capacity without being time-sensitive. - A simplified *long-term consequence module* links reporting delay to the probability of a materially delayed diagnosis. Each material delay is assigned an expected QALY loss and downstream treatment cost. In a full evaluation, this scalar consequence would normally be replaced by a disease-specific state-transition or patient-level model. - Deterministic sensitivity analysis, probabilistic sensitivity analysis (PSA), subgroup analysis, threshold analysis, and expected value of perfect information (EVPI) examine whether uncertainty could change the decision.

Component	What it represents	Principal output
Queue simulation	Arrival variability, reading capacity, downtime, and priority ordering	Time-to-report distributions for all studies and for priority, routine, cancer, and non-cancer groups
Decision tree	Priority-flag performance and immediate consequences	Expected true-positive, false-negative, false-positive, and true-negative counts
Consequence module	Relationship between delay and lifetime consequences	Material delays, QALY loss avoided, and downstream costs avoided
Economic module	Programme costs, offsets, and health outcomes	Incremental cost, ICER, and net monetary benefit

Component	What it represents	Principal output
Uncertainty module	Parameter and structural uncertainty	Sensitivity results, probability of cost-effectiveness, EVPI, and research priorities

Parameters and Assumptions

The following table records the base-case values and the ranges used in deterministic or probabilistic analysis. The values are deliberately transparent so that readers can reproduce the calculations and identify which assumptions should be replaced by local evidence.

Table: Synthetic parameter table

Parameter	Base value	Sensitivity or PSA specification	Role in the model
Annual examinations	25,000	Fixed in base PSA	Service volume
Working days	250	Fixed	Converts annual volume to daily arrivals
Cancer prevalence	0.008	Beta(80, 9920)	Number of time-sensitive cases
Priority sensitivity	0.90	Beta(180, 20); DSA 0.80–0.95	Probability that a cancer case receives priority
Priority specificity	0.92	Beta(920, 80); DSA 0.88–0.96	Controls false-priority volume
Mean daily arrivals	100	Poisson arrivals	Queue demand
Mean daily reading capacity	101.4	Normal mean 101.4, SD 7, truncated at zero	Queue service capacity
Capacity-loss days	5%	25% capacity reduction on affected days	Operational disruption
Baseline mean cancer report time	5.0 days	Normal mean 5.0, SD 1.0, minimum 0.5	Delay under the comparator
True-positive report time	0.5 day	Normal mean 0.5, SD 0.15, minimum 0.2	Delay for correctly prioritised cancer cases
False-negative report time	5.44 days	Normal mean 5.44, SD 1.0, minimum 0.5	Delay for missed priority cases
QALY loss per material delay	0.10	Normal mean 0.10, SD 0.03, minimum 0.01; DSA 0.05–0.15	Lifetime health consequence
Cost per material delay	GBP 2,500	Normal mean 2,500, SD 750, minimum 250; DSA 1,000–5,000	Downstream treatment consequence
Annual licence	GBP 65,000	Triangular(40,000; 65,000; 90,000)	Recurring programme cost
Annualised integration	GBP 20,000	Normal mean 20,000, SD 4,000, minimum 5,000	Interface and technical support
Monitoring and assurance	GBP 18,000	Normal mean 18,000, SD 3,600, minimum 5,000	Performance, equity, safety, and incident review
Training and service support	GBP 8,000	Normal mean 8,000, SD 2,000, minimum 1,000	Workforce implementation cost
Manual triage cost avoided	GBP 20,000	Triangular(0; 20,000; 40,000)	Operational cost offset
Extra review per false-priority study	1.5 minutes	Normal mean 1.5, SD 0.5, minimum 0.1	Priority-review burden
Review cost per minute	GBP 1.20	Fixed	Converts review burden to cost
Base willingness-to-pay threshold	GBP 30,000 per QALY	Scenario at GBP 50,000	Converts health effects to net monetary benefit

Queue and Capacity Model

The queue model assumes that studies arrive throughout each working day and are read from a pooled worklist. The comparator uses first-in, first-out ordering. The intervention places flagged studies ahead of routine studies while preserving first-in, first-out ordering within each class. Service is non-pre-emptive: a study already being interpreted is not interrupted when a new priority study arrives.

The expected priority proportion is

$$P(\text{priority}) = \pi Se + (1-\pi)(1-Sp),$$

where () is prevalence, (Se) is sensitivity, and (Sp) is specificity. In the base case,

[P() = (0.008)(0.90)+(0.992)(0.08) = 0.0866,]

so approximately 8.7% of examinations enter the priority queue.

The synthetic DES used 200 replications of 1,000 working days. The first 500 days of each replication were discarded as warm-up. Random arrivals, variable daily capacity, and temporary capacity loss were represented explicitly. These settings are intended to demonstrate good simulation practice rather than reproduce a particular service.

Assumption	Worked specification
Arrival process	Poisson arrivals with a mean of 100 examinations per working day
Reading capacity	Normally distributed daily capacity with mean 101.4 and standard deviation 7 examinations
Capacity disruption	A 5% daily probability of a 25% reduction in reading capacity
Comparator discipline	First-in, first-out
Intervention discipline	Priority studies first; first-in, first-out within priority and routine classes
Pre-emption	None
Additional capacity	None; the intervention changes ordering rather than staffing or equipment
Warm-up	First 500 days discarded
Replications	200 independent replications
Primary queue outcomes	Mean and 90th-percentile time to report by cancer status and priority class

The queue results in the following table show the central trade-off. Prioritisation substantially reduces reporting time for cancer cases but slightly increases waiting among routine studies. The overall mean changes little because the intervention redistributes capacity rather than adding capacity.

Table: Illustrative queue-model outputs

Output	Comparator	AI priority	Interpretation
Mean time to report, all studies	5.0 days	5.0 days	Reordering does not create additional capacity
90th percentile, all studies	6.7 days	7.2 days	The upper tail for routine studies becomes slightly longer
Mean time to report, cancer cases	5.0 days	1.0 day	Correct prioritisation produces the main potential benefit
Mean time, true-positive cancer cases	5.0 days	0.5 day	Correctly flagged cases are read first
Mean time, false-negative cancer cases	5.0 days	5.44 days	Missed cases remain in the routine queue and may wait longer
Mean time, routine studies	5.0 days	5.44 days	Priority cases impose an opportunity cost on routine work
Priority share	Not applicable	8.7%	Includes true positives and false positives

Priority Classification and Consequences

For an annual cohort of size (N), the expected classification counts are

$$\begin{aligned} TP &\quad N\pi Se, \\ FN &\quad N\pi(1-Se), \\ FP &\quad N(1-\pi)(1-Sp), \\ TN &\quad N(1-\pi)Sp. \end{aligned}$$

The base-case values produce 180 true positives, 20 false negatives, 1,984 false positives, and 22,816 true negatives. These categories matter because a priority system can create both benefit and opportunity cost even when it does not change the final diagnostic interpretation.

Category	**Expected count**	**Queue consequence**	**Clinical and economic consequence**
True positive	180	Read rapidly in the priority queue	Potential reduction in materially delayed diagnosis and downstream treatment cost
False negative	20	Remains in the routine queue	Principal safety concern; delayed interpretation may reduce health benefit
False positive	1,984	Uses priority-reading capacity	Additional review burden and longer routine waiting, without direct diagnostic benefit
True negative	22,816	Remains in the routine queue	Small opportunity cost from reordering; no direct diagnostic benefit

Time to Diagnosis and Long-Term Consequences

A fully developed evaluation would estimate disease-specific relationships among reporting time, diagnostic confirmation, treatment initiation, stage, survival, quality of life, and cost. To keep the worked example transparent, a synthetic relationship is used instead. The probability that a cancer case experiences a materially consequential delay is

$$P_{delay}(d) = min(0.20, 0.04 + 0.02\, max(0, d-1)),$$

where (d) is mean time to report in working days. The function is pedagogical; it is not a clinically validated delay model.

Time to report	**Material-delay probability**	**Interpretation**
1 day or less	0.04	Residual delay remains possible after rapid reporting
3 days	0.08	Moderate increase in expected delay consequence
5 days	0.12	Base-case comparator region
7 days	0.16	Greater likelihood of lost clinical benefit
9 days or more	0.20	Probability capped in the illustrative function

Expected materially delayed cancer cases are

$$D_0 = N\pi P_{delay}(d_0)$$

for the comparator and

$$D_1 = N\pi\left[Se\, P_{delay}(d_{TP}) + (1-Se)\, P_{delay}(d_{FN})\right)$$

for the intervention. Substitution of the base-case values gives

[D_0 = (25,000)(0.008)(0.1202) = 24.04,]

and

[D_1 = (25,000)(0.008) = 9.78.]

The model therefore estimates 14.26 materially delayed diagnoses avoided in one annual cohort.

Costs and Outcomes

If (q) is the QALY loss per material delay and (c_d) is the associated downstream cost, the incremental health effect is

$$\Delta E = \left(D_0 - D_1\right) q .$$

With (q=0.10), the base-case gain is

[E = (24.04-9.78)(0.10) = 1.43 .]

False-priority review costs are calculated as

[C_{FP} = (1,984)(1.5)(1.20) = 3,571.]

The incremental cost is

$$\Delta C = C_{licence} + C_{integration} + C_{monitoring} + C_{training} + C_{FP} - S_{manualtriage} - \left(D_0 - D_1\right) c_d .$$

The base-case calculation is

[C = 65,000+20,000+18,000+8,000+3,571 -20,000-35,657 = 58,914.]

The incremental cost-effectiveness ratio is

$$ICER = \frac{\Delta C}{\Delta E} = \frac{58,914}{1.426} = GBP\ 41,305 \text{ per QALY} .$$

Net monetary benefit at threshold (_{}) is

$$NMB = \lambda_{WTP} \Delta E - \Delta C .$$

At GBP 30,000 per QALY, NMB is approximately GBP -16,125; at GBP 50,000 per QALY, it is approximately GBP 12,401. The base case is therefore not attractive at the lower threshold but becomes favourable at the higher threshold.

Output	Result	Interpretation
Materially delayed diagnoses, comparator	24.04	Expected count in one annual cohort
Materially delayed diagnoses, intervention	9.78	Weighted across true-positive and false-negative pathways
Material delays avoided	14.26	Principal clinical benefit mechanism
Incremental QALYs	1.43	Lifetime consequence assigned to the annual cohort
Gross programme cost	GBP 114,571	Licence, integration, monitoring, training, and false-priority review
Manual triage cost avoided	GBP 20,000	Operational offset
Downstream cost avoided	GBP 35,657	Consequence of fewer material delays
Net incremental cost	GBP 58,914	Gross cost less offsets
ICER	GBP 41,305 per QALY	Above a GBP 30,000 threshold; below a GBP 50,000 threshold

Output	Result	Interpretation
NMB at GBP 30,000 per QALY	GBP -16,125	Unfavourable base-case decision
NMB at GBP 50,000 per QALY	GBP 12,401	Favourable but uncertain base-case decision

Subgroup Assumptions and Equity

Average performance can conceal unequal access, different case complexity, and unequal waiting. Two synthetic subgroups are used to demonstrate this problem. They are not proxies for any protected characteristic and should be replaced by locally relevant clinical, demographic, geographic, disability, language, or access groups.

Table: Synthetic subgroup assumptions and outputs

Measure	Group A	Group B	Difference	Why it matters	Decision response
Share of examinations	80%	20%	–	Different service volume	Preserve subgroup reporting
Cancer prevalence	0.7%	1.2%	+0.5 points	Higher expected need in Group B	Use subgroup-specific prevalence
Priority sensitivity	92%	82%	-10 points	More false negatives in Group B	Investigate data, threshold, and workflow causes
Priority specificity	93%	88%	-5 points	More false priorities in Group B	Monitor review burden and calibration
Baseline cancer report time	4.2 days	6.2 days	+2.0 days	Existing access difference	Do not interpret equal average benefit as equity
AI cancer report time	0.84 day	1.88 days	+1.04 days	Residual disparity remains after prioritisation	Set subgroup time-to-report conditions
False priorities per 1,000 studies	69.5	118.6	+49.1	Greater capacity burden in Group B	Test alternative thresholds and capacity
QALYs gained per 1,000 studies	0.041	0.100	+0.059	Higher potential benefit reflects higher need and longer baseline delay	Report absolute and relative effects together

The example shows that a group can gain more in absolute terms yet still experience poorer performance and longer reporting times. Equity assessment should therefore consider baseline disadvantage, model performance, access to follow-up, absolute benefit, false-priority burden, and residual disparity rather than relying on one average metric.

Deterministic Sensitivity Analysis

One-way sensitivity analysis changes one assumption at a time while holding the others at their base values. The base-case NMB at GBP 30,000 per QALY is GBP -16,125. The following table identifies the parameters most capable of changing that conclusion.

Table: Deterministic sensitivity analysis: net monetary benefit at GBP 30,000 per QALY

Parameter	Low value	High value	NMB at low	NMB at high
Annual licence	GBP 40,000	GBP 90,000	GBP 8,875	GBP -41,125
Priority sensitivity	0.80	0.95	GBP -25,900	GBP -11,238
Priority specificity	0.88	0.96	GBP -17,911	GBP -14,339
Baseline report time	3.0 days	7.0 days	GBP -60,346	GBP 27,654
QALY loss per material delay	0.05	0.15	GBP -37,519	GBP 5,270
Cost per material delay	GBP 1,000	GBP 5,000	GBP -37,519	GBP 19,532

Parameter	Low value	High value	NMB at low	NMB at high
Manual triage savings	GBP 0	GBP 40,000	GBP -36,125	GBP 3,875

The result is most sensitive to baseline delay, the clinical and economic consequences assigned to delay, programme price, and whether existing manual triage costs are genuinely avoided. Sensitivity and specificity matter, but their influence is smaller within the ranges examined. This pattern directs attention to local workflow and outcome evidence rather than to model accuracy alone.

Probabilistic Sensitivity Analysis

The PSA jointly samples uncertain parameters and recalculates costs, QALYs, and NMB. The illustrative analysis used 200,000 Monte Carlo samples. Correlation among parameters was not represented, which is a limitation; local analysis should model known dependence among prevalence, performance, delay, costs, and outcomes. Transparent reporting of AI-specific economic evaluation and value-of-information analysis should follow appropriate reporting guidance (3).

Output	Mean	95% simulation interval
Incremental cost	GBP 59,096	GBP 14,894 to GBP 97,789
Incremental QALYs	1.421	0.416 to 2.832
NMB at GBP 30,000 per QALY	GBP -16,461	GBP -74,216 to GBP 54,628
NMB at GBP 50,000 per QALY	GBP 11,962	GBP -62,610 to GBP 107,281
Probability cost-effective at GBP 30,000 per QALY	28.8%	–
Probability cost-effective at GBP 50,000 per QALY	57.8%	–

The PSA does not support a confident unrestricted-adoption decision. At the lower threshold, most simulations favour the comparator. At the higher threshold, the intervention is more likely than not to be cost-effective, but uncertainty remains substantial.

Value of Information

For strategies (j), the expected value of perfect information is

$$EVPI = E_{\theta}\left[\max_{j} NMB_j(\theta)\right] - \max_{j} E_{\theta}\left[NMB_j(\theta)\right].$$

EVPI is the maximum expected benefit of eliminating all current parameter uncertainty before deciding. It is not the amount that should automatically be spent on research; research cost, delay, feasibility, and the number of future patients affected must also be considered. Reporting should follow transparent value-of-information guidance (4).

Threshold	Population EVPI	EVPI per examination	Interpretation
GBP 30,000 per QALY	GBP 6,817	GBP 0.27	Limited value for an expensive definitive study serving only one annual cohort
GBP 50,000 per QALY	GBP 11,468	GBP 0.46	Greater value because the decision is closer to equipoise

The most useful additional evidence would concern the local relationship between reporting delay and patient outcomes, priority performance in clinically and socially relevant subgroups, queue behaviour during capacity disruption, and the extent to which manual triage work is truly removed. Licence price is influential, but it is primarily a negotiation variable rather than a research parameter. A low-cost silent or controlled pilot may therefore be more proportionate than a large new accuracy study.

Interpretation

The hybrid model reaches four main conclusions. - Prioritisation can reduce reporting delay for cancer cases without changing total reading capacity, but the gain is partly financed by longer waiting among routine studies. - False negatives are the principal clinical risk, whereas false positives create review burden and an opportunity cost for the routine queue. - The base case is not cost-effective at GBP 30,000 per QALY and is only moderately favourable at GBP 50,000 per QALY. Price, baseline delay, delay-related consequences, and realised operational savings are more decision-critical than small changes in specificity. - Average results are insufficient. The subgroup example shows that higher absolute benefit may coexist with poorer sensitivity, more false priorities, and longer residual delay.

The model is intentionally simplified. It omits disease-stage transitions, treatment-capacity constraints, diagnostic anxiety, reader learning, model drift, material updates, environmental costs, and correlations among uncertain parameters. These omissions should be addressed when they could change the institutional decision. Credible modelling requires explicit context of use, version control, verification, validation, limitations, and independent review proportionate to consequence (5).

Recommendation and Decision Conditions

The worked result does not support immediate unrestricted adoption at the base-case price and a GBP 30,000-per-QALY threshold. A controlled local pilot is more defensible because the intervention may create value if local delays are substantial, subgroup performance is acceptable, manual triage costs are removed, and price exposure is reduced.

At the base-case assumptions, the maximum annual licence compatible with zero NMB at GBP 30,000 per QALY is approximately GBP 48,875. At GBP 50,000 per QALY, the corresponding threshold is approximately GBP 77,401. These values are negotiation thresholds derived from the worked assumptions, not market prices.

Domain	Condition	Decision response if not met
Clinical performance	Local sensitivity and calibration are acceptable for the intended pathway and relevant subgroups	Recalibrate, change threshold, restrict use, or stop
Operational performance	Mean cancer reporting time is no more than 1.5 days, while routine mean and 90th-percentile delay remain within agreed limits	Add capacity, redesign the queue, restrict priority volume, or stop
Safety	Every false-negative cancer case and any materially delayed diagnosis is reviewed	Investigate, contain, revise threshold, and reassess intended use
Equity	Subgroup sensitivity, false-priority burden, access to follow-up, and time to report meet predefined conditions	Mitigate, redesign, restrict, or do not scale
Economic value	Price and realised offsets produce non-negative NMB at the institution’s approved threshold	Renegotiate, reduce scope, or do not adopt
Governance	Human interpretation, audit logs, version control, incident response, rollback, and named accountability remain operational	Pause use until controls are restored
Evidence review	Results are reviewed after six and twelve months using a locked protocol and archived model version	Continue only with documented approval and conditions
Exit	Data export, workflow restoration, contract termination, and communication arrangements are tested before dependency increases	Do not scale until exit capability is credible

The final decision should be recorded as one of the following: do not proceed; redesign and repeat; conduct a controlled pilot; adopt with conditions; scale; restrict; pause; replace; or retire. The recommendation should identify the model version, evidence cut-off, accountable owner, unresolved uncertainty, next review date, and stop criteria.

Recommended Companion Files

For a reproducible digital release, this LaTeX narrative should be distributed with: - a parameter workbook in XLSX and ODS formats; - synthetic input data in CSV and JSON formats; - a standard Python script and a Jupyter notebook implementing the queue, economic, subgroup, PSA, and EVPI calculations; - machine-readable model outputs and a static decision summary; - a data dictionary, model specification, test log, software-environment file, random seed, and version history; and - a blank institutional adaptation template that separates local evidence from the synthetic worked values.

Supplemental Resources

These supplemental resources translate the Chapter 8 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 8 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Model Specification	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Verification, Validation, and Reproducibility	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Uncertainty and Value-of-Information Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Model Specification

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Verification, Validation, and Reproducibility

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Specify the comparator, human reading workflow, priority threshold, queue discipline, and downstream pathway. - Use a classification decision tree to estimate true-positive, false-negative, false-positive, and true-negative consequences. - Use discrete-event simulation or a queueing model to estimate time-to-report distributions under constrained reader capacity. - Link material diagnostic delay to downstream cost and QALY consequences. - Represent subgroup prevalence, performance, access, and delay distributions explicitly. - Conduct deterministic sensitivity analysis, probabilistic sensitivity analysis, and value-of-information analysis.

Uncertainty and Value-of-Information Record

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 8

Term	Working definition and decision relevance
Decision model	An explicit simplification of a decision problem and its causal and operational structure.
Simulation	Computational representation of events, histories, queues, actors, or feedback.
Value of information	The expected benefit of reducing uncertainty before or during commitment.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 8

Instrument	Status or type	Appropriate use and limitation
Decision tree	Model structure	Suitable for short, nonrecurrent pathways.
State-transition model	Model structure	Represents repeated movement among health states.
Microsimulation	Individual-level model	Represents heterogeneity and history.
DES	Simulation method	Represents queues, resources, events, and timing.
ABM/system dynamics	Simulation methods	Represent interactions or aggregate feedback.
CHEERS-VOI	Reporting guideline	Supports reporting of value-of-information analysis.

Formula and Interpretation Note

The following relationship is used in Chapter 8 or its companion resources:

$$EVPI = E_{\theta}\left[\max_{j} NMB_{j}(\theta)\right) - \max_{j} E_{\theta}\left[NMB_{j}(\theta)\right)$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE08 — Imaging Prioritization Decision Model and Value of Information Worked case; md; csv; json; xlsx; ods	Which uncertainty is most likely to change the adoption decision, and is further evidence worth its cost?
AIHE-B05 — Model Selection, Uncertainty, and Reproducibility Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Select a decision model from the problem structure and document evidence, verification, uncertainty, and reproducibility.
AIHE-RG05 — Scenario Planning and Real-Options Worksheet Reference guide; docx; odt; xlsx; ods; csv; json; md	Examine uncertainty through alternative futures, thresholds, staged commitments, learning options, contract flexibility, and explicit exit criteria.
AIHE-PY14 — Deterministic sensitivity and tornado analysis Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Tests how one uncertain input at a time changes net monetary benefit or another linear decision outcome and ranks decision drivers.
AIHE-PY15 — Probabilistic sensitivity analysis, CEAC, and EVPI Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Propagates uncertainty in incremental costs and effects through Monte Carlo simulation, estimates cost-effectiveness probabilities across thresholds, and computes expected value of perfect information.
AIHE-PY18 — Decision-tree pathway model Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Calculates expected cost, effect, events, and net monetary benefit for short AI and comparator pathways with mutually exclusive outcome branches.
AIHE-PY19 — Markov cohort model for long-term AI evaluation Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Simulates repeated transitions among health states for AI and usual care, accumulating discounted costs and QALYs over time.
AIHE-PY20 — Patient-level microsimulation Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Simulates heterogeneous patients over repeated cycles so history, subgroup risk, adherence, and individual outcomes can affect AI value.
AIHE-AF — Evidence, Reporting, and Assurance Crosswalk Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Match evidence and assurance needs to reporting standards, appraisal tools, lifecycle guidance, and management systems.
AIHE-EX04 — Evidence and Assumption Log Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Link each material claim and assumption to evidence, quality, applicability, uncertainty, owner, and action.
AIHE-EX06 — Model Card and AI Intervention Fact Sheet Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd;	Record the exact system, version, intended and prohibited uses, evidence, limitations, human oversight, and monitoring.

Resource	Purpose
svg; png	
AIHE-EX15 — Analytic and Simulation Method Selection Guide Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Select decision trees, Markov models, microsimulation, discrete-event simulation, system dynamics, or agent-based modeling by problem structure.
AIHE-EX16 — Reproducible Analysis, Model Risk, and Code Quality Checklist Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Verify data provenance, environments, versioning, testing, validation, uncertainty, security, release, and archive controls.
AIHE-LTX-08-REF — Reference Resources nan; TeX	Chapter 8 reference resource aligned with the current book structure.
AIHE-LTX-08-SUP — Supplemental Resources nan; TeX	Chapter 8 supplemental resource aligned with the current book structure.
AIHE-LTX-08-WOR — Imaging Prioritisation Decision Model nan; TeX	Chapter 8 worked-example resource aligned with the current book structure.

Chapter 9: Problem-First Procurement and Vendor Governance

Chapter Orientation

Chapter 9 translates the problem, evidence, economics, safeguards, and implementation requirements into procurement and continuing vendor governance. The resources support pre-procurement specification, due diligence, non-compensatory safeguards, weighted comparison, update control, audit rights, pricing, continuity, renegotiation, and exit.

Use this suite before tendering, during vendor evaluation, at contract negotiation, and when renewal or renegotiation is required. The records are designed to reduce information asymmetry and to preserve local evidence access, monitoring, portability, rollback, and decommissioning options.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Vendor Procurement and Renegotiation	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Pre-Procurement Requirements; Vendor Scorecard and Mandatory Safeguards; Contract, Renegotiation, and Exit Record	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Problem-first procurement, Mandatory safeguard, Vendor governance	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	1 core, 2 python analytics, 3 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Vendor Procurement and Renegotiation

Purpose and Status

A hospital is procuring a clinically consequential AI platform. Three vendors differ in evidence quality, interoperability, pricing, monitoring access, update control, and exit support. A prior procurement also requires renegotiation because implementation costs and vendor restrictions exceeded the original assumptions.

Decision Problem and Comparators

Decision problem. Which vendor, if any, should proceed to a controlled contract, and what remedies or exit actions are required for the existing platform?

Table: Comparators for the Chapter 9 worked example

Option	Comparator or strategy
1	Vendor A: strong evidence, high fixed price
2	Vendor B: lower price, limited monitoring access
3	Vendor C: shared-service model with stronger exit terms
4	Internal or non-AI process alternative
5	Renegotiate, replace, restrict, or retire the existing platform

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$S_v = \sum_{d=1}^{D} w_d s_{vd}, G_v = \prod_{k=1}^{K} g_{vk}$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 9 worked example

Parameter	Base value	Source or interpretation
Mandatory safeguard domains	7	Procurement specification
Weighted domains	7	Local scorecard
Expected annual volume	180,000 uses	Activity forecast
Maximum acceptable variable exposure	EUR 420,000/year	Budget condition
Required update notice	90 days	Contract requirement
Minimum data-export period after termination	12 months	Continuity requirement
Remediation period for existing vendor	120 days	Renegotiation proposal

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Define the service problem, comparator set, intended and excluded uses, and measurable outcomes.	Evidence and responsible owner to be recorded
2	Require version-matched evidence and complete technical, data, security, economic, and workflow due diligence.	Evidence and responsible owner to be recorded
3	Apply mandatory safeguards before weighted comparison.	Evidence and responsible owner to be recorded
4	Model pricing and lifecycle cost under realistic volume, update, support, and exit scenarios.	Evidence and responsible owner to be recorded
5	Convert monitoring, audit, update, continuity, portability, and exit requirements into enforceable clauses.	Evidence and responsible owner to be recorded
6	For the existing platform, compare prospective continuation, renegotiation, restriction, replacement, and decommissioning without allowing sunk cost to determine the decision.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Vendor B weighted score	Highest before safeguards	Ineligible because monitoring access fails a mandatory requirement
Vendor A evidence profile	Strongest	Price and exit terms require negotiation
Vendor C continuity profile	Strongest	Local evidence and governance arrangements require confirmation
Existing vendor decision	Time-limited remediation with parallel exit plan	Continuation is conditional
Preferred award approach	Competitive dialogue followed by conditional	No automatic award from aggregate score

Output	Result	Interpretation
	contract	

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Proceed only with a vendor that passes every mandatory safeguard and accepts version-specific evidence, monitoring access, transparent pricing, material-change control, continuity, and exit obligations. Continue the existing platform only under a time-limited remediation agreement and maintain a funded replacement plan.

- mandatory safeguards are evaluated independently of weighted preferences.
- missing evidence receives clarification, conservative treatment, or exclusion.
- contracted pricing includes integration, monitoring, updates, incident support, and exit.
- the institution retains audit, data, log, and publication rights needed for accountability.
- renewal is linked to realized value and current evidence rather than sunk cost.

Supplemental Resources

These supplemental resources translate the Chapter 9 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 9 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Pre-Procurement Requirements	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Vendor Scorecard and Mandatory Safeguards	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Contract, Renegotiation, and Exit Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Pre-Procurement Requirements

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Vendor Scorecard and Mandatory Safeguards

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define the service problem, comparator set, intended and excluded uses, and measurable outcomes. - Require version-matched evidence and complete technical, data, security, economic, and workflow due diligence. - Apply mandatory safeguards before weighted comparison. - Model pricing and lifecycle cost under realistic volume, update, support, and exit scenarios. - Convert monitoring, audit, update, continuity, portability, and exit requirements into enforceable clauses. - For the existing platform, compare prospective continuation, renegotiation, restriction, replacement, and decommissioning without allowing sunk cost to determine the decision.

Contract, Renegotiation, and Exit Record

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 9

Term	Working definition and decision relevance
Problem-first procurement	A process that defines need, comparator, evidence, and safeguards before product selection.
Mandatory safeguard	A non-compensatory requirement that must be met before weighted comparison.
Vendor governance	Continuing oversight of evidence, performance, change, continuity, and exit after award.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 9

Instrument	Status or type	Appropriate use and limitation
NICE Evidence Standards Framework	Evidence guidance	Supports evidence planning for digital health technologies.
DTAC	Jurisdiction-specific assessment criteria	Supports baseline digital-technology due diligence in England.
EU model contractual clauses	Public procurement resource	Provides adaptable AI contractual clauses.
FAIR-AI	Implementation and review framework	Links pre-implementation review with post-implementation monitoring.

Formula and Interpretation Note

The following relationship is used in Chapter 9 or its companion resources:

$$S_v = \sum_{d=1}^{D} w_d s_{vd}, G_v = \prod_{k=1}^{K} g_{vk}$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE09 — AI Vendor Procurement and Contract Negotiation Worked case; md; csv; json; xlsx; ods	How should the hospital compare vendors when a lower price is accompanied by weak audit, security, update, and exit provisions?
AIHE-D01 — Vendor Evidence and Contract Requirements Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Translate evidence, interoperability, data, security, monitoring, update, service, remedy, portability, and exit requirements into contract terms.
AIHE-PY17 — AI pricing and contract scenario analysis Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Compares subscription, per-use, risk-sharing, and hybrid pricing arrangements under different volume, uptake, performance, and savings assumptions.
AIHE-PY25 — Weighted procurement scorecard with mandatory safeguards Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Ranks candidate AI products using weighted evidence domains while enforcing non-compensatory safety, privacy, equity, cybersecurity, audit, and exit safeguards.
AIHE-D — AI Procurement Checklist Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Evaluate problem fit, evidence, integration, lifecycle cost, data rights, security, monitoring, updates, audit, and exit.
AIHE-EX08 — Vendor Demonstration Evaluation Form Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd;	Structure demonstrations around real workflows, evidence, limitations, usability, interoperability, monitoring, and unsupported claims.

Resource	Purpose
svg; png	
AIHE-EX09 — Contract Requirements and Exit Matrix Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Document intended use, evidence, audit, updates, data, security, service levels, pricing, termination, portability, and decommissioning clauses.
AIHE-LTX-09-REF — Reference Resources nan; TeX	Chapter 9 reference resource aligned with the current book structure.
AIHE-LTX-09-SUP — Supplemental Resources nan; TeX	Chapter 9 supplemental resource aligned with the current book structure.
AIHE-LTX-09-WOR — Vendor Procurement and Renegotiation nan; TeX	Chapter 9 worked-example resource aligned with the current book structure.

Part IV: Governance, Evidence, and Lifecycle Decision-Making

This Part brings governance, accountability, evidence, validation, lifecycle operations, and continuing decision-making into one institutional system.

Chapter 10: Governance, Accountability, and Institutional Risk

Chapter Orientation

Chapter 10 integrates governance, accountability, privacy, cybersecurity, equity, digital inclusion, trust, and institutional risk. The resources make decision rights explicit and prevent favourable expected value from compensating for an unacceptable safeguard failure.

Use this suite for patient-facing systems, consequential decision support, public communication, and any intervention whose risks extend beyond technical performance. The outputs should identify affected groups, lawful authority, meaningful human routes, accessibility, incident response, residual risk, and the authority to restrict or stop use.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Patient-Facing Chatbot and Digital Inclusion	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Governance and Accountability Record; Integrated Institutional-Risk Register; Patient-Facing AI Assurance Record	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Governance, Accountability, Institutional risk	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	2 core, 3 python analytics, 1 reference guide, 6 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Patient-Facing Chatbot and Digital Inclusion

Purpose and Status

A hospital proposes a multilingual patient-facing chatbot for appointment guidance, preparation instructions, and symptom navigation. The intervention could improve access but may also create factual, privacy, cybersecurity, accessibility, workload, and trust risks.

Decision Problem and Comparators

Decision problem. Should the chatbot be piloted, restricted to administrative tasks, redesigned with human escalation, or rejected in favour of non-AI navigation improvements?

Table: Comparators for the Chapter 10 worked example

Option	Comparator or strategy
1	Improved static web and printed information
2	Call-centre redesign and multilingual staffing
3	Rules-based conversational interface
4	Restricted AI chatbot with human escalation

Model and Decision Structure

A compact quantitative relationship used in this example is:

No compensatory risk equation is recommended; mandatory safeguards are assessed as gates.

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 10 worked example

Parameter	Base value	Source or interpretation
Languages proposed	8	Service plan
Share of users requiring assisted digital access	18%	Patient survey
Queries classified as clinically consequential	24%	Retrospective sample
Human escalation target	Under 5 minutes	Safety requirement
Retention of interaction logs	30 days	Draft policy
Expected annual interactions	420,000	Activity forecast

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Define allowed, prohibited, and escalation-required tasks.	Evidence and responsible owner to be recorded
2	Map data flows, third parties, retention, logging, and model-service dependencies.	Evidence and responsible owner to be recorded
3	Assess factuality, information integrity, privacy, cybersecurity, language, accessibility, and digital inclusion.	Evidence and responsible owner to be recorded
4	Estimate the effect on call-centre demand, repeat contacts, abandonment, and downstream services.	Evidence and responsible owner to be recorded
5	Assign accountability, incident communication, contestability, and stopping authority.	Evidence and responsible owner to be recorded
6	Pilot only where essential non-digital routes remain available and monitoring can detect differential use and harm.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Suitable initial scope	Administrative guidance and verified preparation instructions	Symptom advice remains restricted
Primary equity risk	Exclusion of users with language, disability, literacy, connectivity, or trust barriers	Requires human alternatives
Primary governance risk	Unclear responsibility for harmful or incomplete advice	Requires explicit escalation and logs
Expected call reduction	Uncertain	Must account for repeat and escalated contacts
Preferred decision	Restricted pilot with independent content verification	Not unrestricted patient-facing deployment

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Pilot a restricted, clearly disclosed service for low-consequence tasks, preserve immediate human alternatives, and prohibit unsupported diagnosis or treatment advice. Expansion should depend on effective access and outcome evidence rather than interaction volume.

- all content domains have accountable clinical or operational owners.
- language and accessibility testing includes affected users.
- privacy, retention, and third-party data use are transparent and minimized.
- escalation reliability and abandonment are monitored by subgroup.
- a serious factual, privacy, security, or equity failure triggers immediate restriction and review.

Supplemental Resources

These supplemental resources translate the Chapter 10 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 10 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Governance and Accountability Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Integrated Institutional-Risk Register	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Patient-Facing AI Assurance Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Governance and Accountability Record

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Integrated Institutional-Risk Register

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define allowed, prohibited, and escalation-required tasks. - Map data flows, third parties, retention, logging, and model-service dependencies. - Assess factuality, information integrity, privacy, cybersecurity, language, accessibility, and digital inclusion. - Estimate the effect on call-centre demand, repeat contacts, abandonment, and downstream services. - Assign accountability, incident communication, contestability, and stopping authority. - Pilot only where essential non-digital routes remain available and monitoring can detect differential use and harm.

Patient-Facing AI Assurance Record

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 10

Term	Working definition and decision relevance
Governance	Institutional decisions, authority, controls, records, and responses governing consequential use.
Accountability	The obligation to justify decisions and accept responsibility for consequences.
Institutional risk	Risk to safety, rights, continuity, equity, trust, resources, and legitimacy.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 10

Instrument	Status or type	Appropriate use and limitation
EU AI Act	Binding EU regulation in scope	Creates obligations based on role and risk classification.
EHDS	Binding EU regulation with phased implementation	Governs electronic health data use and infrastructure in scope.
ISO/IEC 42001, 23894, 42005	Voluntary international standards	Support management systems, risk management, and impact assessment.
NIST AI RMF	Voluntary risk-management framework	Organizes Govern, Map, Measure, and Manage activities.
FUTURE-AI and FAIR-AI	Research and consensus guidance	Support trustworthy implementation and review; not regulatory standards.

Formula and Interpretation Note

The following relationship is used in Chapter 10 or its companion resources:

No compensatory risk equation is recommended; mandatory safeguards are assessed as gates.

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE10 — Patient-Facing Chatbot Institutional-Risk Review Worked case; md; csv; json; xlsx; ods	Which uses can be permitted when convenience and call reduction coexist with privacy, safety, equity, and communication risks?
AIHE-C03 — Integrated Institutional-Risk Review Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Review ethics, equity, privacy, cybersecurity, safety, continuity, trust, legal, and economic risks as one institutional system.
AIHE-C05 — Public Transparency and Communication Record Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Define accurate, accessible communication about intended use, limitations, oversight, data, complaints, incidents, and alternatives.
AIHE-RG04 — Regulatory, Standards, and Policy Update Register Reference guide; docx; odt; xlsx; ods; csv; json; md	Maintain date-stamped records for European, Romanian, and global legal, policy, standards, reimbursement, and guidance sources whose scope or implementation status may change.
AIHE-PY07 — Feature importance and partial-dependence exploration Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Fits a tree ensemble, calculates permutation importance, and visualizes the relationship between the top feature and predicted risk.
AIHE-PY08 — Subgroup performance and equity audit Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Compares AUC, calibration, sensitivity, specificity, predictive values, and selection rates across population groups.
AIHE-PY26 — Integrated privacy, cybersecurity, equity, safety, and trust risk register Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Scores inherent and residual risk, checks mitigation completeness, prioritizes overdue actions, and produces an accountable risk dashboard.
AIHE-AB — Privacy and Cybersecurity Risk Register Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Record privacy, security, continuity, supplier, model, and incident risks with controls, owners, and review dates.

Resource	Purpose
AIHE-AC — Equity Impact Tool Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Identify affected groups, barriers, harms, mitigations, costs, residual concerns, and monitoring indicators.
AIHE-C — Ethical Governance Checklist for Healthcare AI Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Translate autonomy, safety, transparency, fairness, oversight, accountability, and public value into evidence and action.
AIHE-EX13 — Public Transparency and Communication Notice Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Explain AI use, limits, human oversight, data handling, accessibility, questions, complaints, incidents, and human alternatives.
AIHE-G — Sample Health-System AI Governance Charter Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Establish a health-system governance charter covering purpose, scope, membership, evidence, decisions, monitoring, incidents, updates, and de-adoption.
AIHE-N — Trust, Equity, and Public Value Workbook Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Integrate trust-adjusted value, distributional effects, mitigation, public value, confidence, and mandatory safeguards.
AIHE-LTX-10-REF — Reference Resources nan; TeX	Chapter 10 reference resource aligned with the current book structure.
AIHE-LTX-10-SUP — Supplemental Resources nan; TeX	Chapter 10 supplemental resource aligned with the current book structure.
AIHE-LTX-10-WOR — Patient-Facing Chatbot and Digital Inclusion nan; TeX	Chapter 10 worked-example resource aligned with the current book structure.

Chapter 11: Evidence, Validation, and Real-World Evaluation

Chapter Orientation

Chapter 11 addresses evidence maturity, appraisal, external and local validation, silent evaluation, causal evaluation, implementation research, mixed methods, and evidence synthesis. The resources distinguish model validity from the effect of using the complete intervention in practice.

Use this suite to design local validation, prospective evaluation, subgroup analysis, decision-curve analysis, causal studies, and decision-ready evidence records. The connected readmission case separates predictive performance from the effect of the care-management pathway and from the resources needed to deliver it.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Readmission-Risk Prediction: Validation and Causal Evaluation	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Evidence Maturity and Appraisal; Local Validation and Silent Evaluation; Causal and Real-World Evaluation Protocol	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Validation, Clinical utility, Causal evaluation	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	6 python analytics, 2 reference guide, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Readmission-Risk Prediction: Validation and Causal Evaluation

Purpose and Status

A hospital has acquired a readmission-risk model intended to direct care-management follow-up. External validation is favourable, but the local population, documentation, social-risk information, and care-management capacity differ from the development setting.

Decision Problem and Comparators

Decision problem. Is the exact model and intervention sufficiently valid and useful for local use, and does the targeted follow-up pathway cause better outcomes than current discharge planning?

Table: Comparators for the Chapter 11 worked example

Option	Comparator or strategy
1	Current discharge planning
2	Conventional risk checklist
3	Model-guided follow-up with existing capacity
4	Model-guided follow-up with expanded care-management capacity

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$BS = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{p}_i - y_i\right)^2$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 11 worked example

Parameter	Base value	Source or interpretation
Annual eligible discharges	18,000	Hospital activity
30-day readmission prevalence	14%	Baseline cohort
Available care-management slots	65/week	Service capacity
Model sensitivity at proposed threshold	0.76	Local silent evaluation
Positive predictive value	0.31	Local silent evaluation
Calibration slope	0.82	Local validation
Expected follow-up effect	Relative risk 0.88	Evidence synthesis; causal uncertainty remains

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Define the intended use, threshold, action, capacity, and exact system version.	Evidence and responsible owner to be recorded
2	Assess data design, leakage, transportability, discrimination, calibration, threshold-specific utility, and subgroup performance.	Evidence and responsible owner to be recorded
3	Conduct silent evaluation before clinical dependence.	Evidence and responsible owner to be recorded
4	Separate model validation from causal evaluation of the follow-up intervention.	Evidence and responsible owner to be recorded
5	Use an appropriate randomized, quasi-experimental, or target-trial design to estimate service effects.	Evidence and responsible owner to be recorded
6	Combine technical, clinical, implementation, qualitative, economic, governance, and post-deployment evidence in a decision-ready record.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Local discrimination	Acceptable	Not sufficient for adoption
Calibration	Overprediction in higher-risk groups	Requires recalibration or threshold review
Weekly patients flagged	Approximately 120	Exceeds current care-management capacity
Causal effect of follow-up	Uncertain locally	Requires controlled evaluation
Preferred decision	Recalibrate and conduct capacity-constrained pilot	No unrestricted implementation

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Complete recalibration and a prospective, capacity-aware evaluation of the model-plus-follow-up intervention. The model should not be judged effective merely because it predicts readmission; the evaluation must show that targeted action improves outcomes relative to realistic alternatives.

- silent-mode evidence and subgroup calibration are acceptable.
- the threshold fits available care-management capacity.
- the causal evaluation separates model performance from intervention effect.
- implementation costs and workload are measured prospectively.
- the evidence record states attribution limits and conditions for continuation.

Supplemental Resources

These supplemental resources translate the Chapter 11 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 11 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Evidence Maturity and Appraisal	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Local Validation and Silent Evaluation	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Causal and Real-World Evaluation Protocol	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Evidence Maturity and Appraisal

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Local Validation and Silent Evaluation

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define the intended use, threshold, action, capacity, and exact system version. - Assess data design, leakage, transportability, discrimination, calibration, threshold-specific utility, and subgroup performance. - Conduct silent evaluation before clinical dependence. - Separate model validation from causal evaluation of the follow-up intervention. - Use an appropriate randomized, quasi-experimental, or target-trial design to estimate service effects. - Combine technical, clinical, implementation, qualitative, economic, governance, and post-deployment evidence in a decision-ready record.

Causal and Real-World Evaluation Protocol

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 11

Term	Working definition and decision relevance
Validation	Assessment of performance and suitability for a stated context of use.
Clinical utility	Whether use of information improves decisions or outcomes relative to alternatives.
Causal evaluation	Estimation of what would have happened under a credible counterfactual.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 11

Instrument	Status or type	Appropriate use and limitation
TRIPOD+AI	Reporting guideline	Prediction-model studies.
PROBAST+AI	Appraisal tool	Risk of bias and applicability for prediction-model evidence.
STARD-AI	Reporting guideline	Diagnostic-accuracy studies.
SPIRIT-AI/CONSORT-AI	Reporting guidelines	Trial protocols and reports.
DECIDE-AI	Reporting guideline	Early-stage live clinical evaluation.
MRC/CFIR/RE-AIM/PRISM	Methodological and implementation frameworks	Support programme theory, determinants, outcomes, context, and sustainment.

Formula and Interpretation Note

The following relationship is used in Chapter 11 or its companion resources:

$$BS = \frac{1}{n} \sum_{i=1}^{n} \left(\hat{p}_i - y_i \right)^2$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE11 — Local Validation and Causal Evaluation of a Readmission Model Worked case; md; csv; json; xlsx; ods	Does local performance and observed service change support adoption, and what can reasonably be attributed to the AI-enabled pathway?
AIHE-RG02 — Reporting and Appraisal Tool Selection Guide Reference guide; docx; odt; xlsx; ods; csv; json; md	Match evidence type and study stage to current reporting and appraisal instruments without treating checklist completion as proof of validity, effectiveness, or regulatory conformity.
AIHE-RG06 — Diagnostic and Predictive Evidence Appraisal Checklist Reference guide; docx; odt; xlsx; ods; csv; json; md	Appraise reference standards, index tests, data partitions, calibration, diagnostic accuracy, bias, subgroup performance, clinical utility, and downstream resource consequences.
AIHE-PY04 — Predictive model development: logistic regression and random forest Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Builds two baseline classifiers, compares test performance, and exports predictions for later validation modules.
AIHE-PY05 — Model performance, threshold trade-offs, and calibration Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Calculates ROC AUC, Brier score, confusion-matrix measures across thresholds, calibration intercept and slope, and calibration bins.
AIHE-PY24 — Implementation evaluation with interrupted time	Evaluates change after an AI implementation using segmented regression,

Resource	Purpose
series and control charts Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	run charts, control limits, uptake, and balancing measures.
AIHE-PY30 — Random-effects meta-analysis and forest plot Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Pools study-level effects using inverse-variance fixed and DerSimonian-Laird random-effects methods, quantifies heterogeneity, and creates forest and funnel plots.
AIHE-PY31 — Difference-in-differences evaluation Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Estimates the effect of AI implementation by comparing change over time in an intervention group with change in a comparison group.
AIHE-PY32 — Time-to-event and survival analysis Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Estimates Kaplan-Meier survival curves and compares event timing between AI and comparator groups using a log-rank test.
AIHE-LTX-11-REF — Reference Resources nan; TeX	Chapter 11 reference resource aligned with the current book structure.
AIHE-LTX-11-SUP — Supplemental Resources nan; TeX	Chapter 11 supplemental resource aligned with the current book structure.
AIHE-LTX-11-WOR — Readmission-Risk Prediction: Validation and Causal Evaluation nan; TeX	Chapter 11 worked-example resource aligned with the current book structure.

Chapter 12: Decision Architecture and Lifecycle Governance

Chapter Orientation

Chapter 12 brings the book's methods into one decision architecture. The resources organize AI-HEVG assessment, the AI Health Economics Dossier, post-deployment monitoring, algorithmovigilance, incidents, material change, staged implementation, renewal, de-adoption, and portfolio governance.

Use this suite when evidence from several specialist analyses must be assembled into an accountable institutional decision and maintained after deployment. The records emphasize continuing responsibility, version control, decision triggers, mandatory safeguards, and a documented route to restriction, replacement, or retirement.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Readmission-Risk Lifecycle Decision	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Integrated Assessment and AI-HED; Post-Deployment Monitoring and Incident Record; Accountable Adoption Principles	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Decision architecture, Algorithmovigilance, AI-HED	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	4 core, 3 python analytics, 1 reference guide, 16 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Readmission-Risk Lifecycle Decision

Purpose and Status

Following local validation and a controlled pilot, the hospital must decide whether to scale the readmission-risk intervention. The evidence dossier includes model performance, care-management outcomes, cost, subgroup effects, workload, vendor updates, and unresolved uncertainty.

Decision Problem and Comparators

Decision problem. Should the intervention scale, continue with conditions, remain restricted, be redesigned, or be de-adopted?

Table: Comparators for the Chapter 12 worked example

Option	Comparator or strategy
1	Continue limited pilot
2	Scale with current threshold and capacity
3	Scale with expanded care-management capacity
4	Restrict to selected services or subgroups
5	Return to the conventional discharge pathway

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$\text{Decision} = f\left(\text{evidence, value, feasibility, safeguards, ownership, reversibility}\right)$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 12 worked example

Parameter	Base value	Source or interpretation
Pilot duration	9 months	AI-HED record
Readmission relative risk	0.91	Local controlled evaluation
Incremental cost per discharge	EUR 94	TCAIO record
Care-management capacity used	93%	Operations dashboard
Subgroup with calibration concern	Patients with incomplete social-risk data	EWAHE review
Material vendor update expected	Within 6 months	Change notice

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Use AI-HEVG to integrate problem relevance, data readiness, clinical validity, economic value, ethics, and organizational feasibility.	Evidence and responsible owner to be recorded
2	Use AI-HED as the living decision record linking evidence, assumptions, decisions, and versions.	Evidence and responsible owner to be recorded
3	Apply non-compensatory safeguards before considering aggregate scores.	Evidence and responsible owner to be recorded
4	Assign accountable roles through HOAM and define monitoring, incidents, material change, renewal, and exit.	Evidence and responsible owner to be recorded
5	Use GATED-AI, VALIDATE-AI, and other processes only for their defined operational roles.	Evidence and responsible owner to be recorded
6	Reach a decision with conditions, monitoring thresholds, review dates, and stopping authority.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Clinical effect	Modest and uncertain	Direction favourable; magnitude below initial expectation
Economic value	Conditional	Dependent on capacity expansion and avoided admissions
Equity concern	Material but potentially remediable	Incomplete social-risk data affect prioritisation
Lifecycle concern	Upcoming material update	Requires version-specific reassessment
Preferred decision	Continue with conditions; do not scale fully	Evidence and capacity not yet sufficient

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Continue the intervention within its current scope while addressing social-risk data, care-management capacity, and the forthcoming material update. Full scale should require renewed validation and a revised economic and equity assessment.

- AI-HED is updated with the exact deployed version, evidence cut-off, and decision rationale.
- monitoring includes workload, outcomes, subgroup access, cost, incidents, and trust.
- the material update is assessed before deployment and rollback remains available.
- the accountable authority can restrict or retire use without vendor approval.
- renewal is based on current evidence and realized value rather than prior investment.

Supplemental Resources

These supplemental resources translate the Chapter 12 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 12 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Integrated Assessment and AI-HED	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Post-Deployment Monitoring and Incident Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Accountable Adoption Principles	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Integrated Assessment and AI-HED

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Post-Deployment Monitoring and Incident Record

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Use AI-HEVG to integrate problem relevance, data readiness, clinical validity, economic value, ethics, and organizational feasibility. - Use AI-HED as the living decision record linking evidence, assumptions, decisions, and versions. - Apply non-compensatory safeguards before considering aggregate scores. - Assign accountable roles through HOAM and define monitoring, incidents, material change, renewal, and exit. - Use GATED-AI, VALIDATE-AI, and other processes only for their defined operational roles. - Reach a decision with conditions, monitoring thresholds, review dates, and stopping authority.

Accountable Adoption Principles

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 12

Term	Working definition and decision relevance
Decision architecture	The integrated arrangement connecting evidence, economics, governance, implementation, records, and action.
Algorithmovigilance	An emerging term for systematic post-deployment evaluation and monitoring of healthcare algorithms.
AI-HED	The proposed living dossier linking the decision problem, evidence, versions, decisions, and lifecycle actions.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 12

Instrument	Status or type	Appropriate use and limitation
AI-HEVG	Author-developed integrated framework	Structures assessment of problem, data, evidence, economics, ethics, feasibility, and lifecycle governance.
AI-HED	Author-developed decision record	Maintains a living versioned dossier.
GATED-AI	Author-developed process	Structures decision gates for procurement and approval.
VALIDATE-AI	Author-developed process	Structures staged implementation.
SHARE-AI	Author-developed process	Structures governed secondary data use.
Algorithmovigilance	Emerging scholarly and operational term	Supports post-deployment monitoring; not a standardized regulatory concept.

Formula and Interpretation Note

The following relationship is used in Chapter 12 or its companion resources:

$$\text{Decision} = f\left(\text{evidence, value, feasibility, safeguards, ownership, reversibility}\right)$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE12 — Model Update, Incident Response, and Renewal Decision Worked case; md; csv; json; xlsx; ods	Should the organization continue, restrict, roll back, replace, or retire the system after a material update and operational incident?
AIHE-D02 — Lifecycle Gates, Staged Implementation, and Dossier Status Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Control increasing commitment through problem, readiness, evidence, local validation, economics, pilot, scale, renewal, and retirement gates.
AIHE-D03 — Algorithmovigilance Operating Cadence Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Define recurring and event-triggered surveillance across technical, workflow, safety, equity, economic, and contractual domains.
AIHE-D04 — Incident Classification and Reconstruction Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Reconstruct technical, clinical, workflow, privacy, cybersecurity, equity, communication, and economic incidents consistently.
AIHE-D05 — Change-Control and Renewal Record Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Assess material changes and periodic renewal across system, data, workflow, vendor, price, law, evidence, alternatives, and exit.
AIHE-RG01 — Framework Status and Selection Guide Reference guide; docx; odt; xlsx; ods; csv; json; md	Distinguish legal instruments, standards, guidance, reporting guidelines, appraisal tools, implementation frameworks, and author-developed

Resource	Purpose
	decision aids; select them according to function and authority.
AIHE-PY09 — Temporal validation and drift monitoring Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Tracks discrimination, calibration error, score distribution, and feature drift across time periods.
AIHE-PY27 — Algorithmovigilance, lifecycle cadence, drift, and incident monitoring Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Monitors technical, clinical, workflow, equity, economic, and trust indicators against warning and stop thresholds over time.
AIHE-PY29 — AI-HED completeness and institutional decision-quality audit Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Checks whether a proposed AI decision dossier covers the problem, comparator, evidence, data, economics, ethics, implementation, monitoring, and exit requirements.
AIHE-AE — AI-HED Completeness and Decision-Quality Review Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Review dossier completeness, confidence, inconsistencies, mandatory safeguards, decision readiness, and remediation.
AIHE-E — Applied Case and Decision Review Template Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Apply the book architecture to an institutional case and produce a defendable decision with conditions and lifecycle controls.
AIHE-EX02 — AI System Inventory and Policy Register Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Maintain a controlled inventory of proposed and deployed AI systems, owners, versions, risk levels, and review dates.
AIHE-EX10 — Workflow Change and Training Competency Log Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Document role, workflow, training, competency, downtime, escalation, refresh, and adoption requirements.
AIHE-EX11 — Material Change and Renewal Review Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Assess model, data, interface, workflow, vendor, contract, policy, performance, risk, and annual renewal changes.
AIHE-EX12 — Incident Corrective Action and Reconstruction Log Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Classify and reconstruct technical, clinical, workflow, privacy, cybersecurity, equity, and communication incidents.
AIHE-EX14 — Final Board Recommendation Pack Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Prepare a decision-ready executive recommendation, options, evidence, economics, residual risk, dissent, conditions, owners, and next review.
AIHE-H — AI Project One-Page Decision Memo Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Summarize the decision problem, comparator, evidence, economics, risks, implementation, conditions, and decision requested.
AIHE-J — Lifecycle and Future-Proofing Toolkit Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Bundle lifecycle checklists for monitoring, version control, scenario planning, policy updates, procurement, and decommissioning.
AIHE-K — AI-HEVG Assessment Workbook Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Assess problem relevance, data, validity, economics, ethics, feasibility, lifecycle governance, safeguards, and final decision.
AIHE-P — AI-HED Dossier Template Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Maintain a versioned evidence, economics, governance, implementation, monitoring, update, and decision record.
AIHE-Q — GATED-AI Gate Record Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Control commitment through problem, evidence, data, economics, governance, implementation, contract, and lifecycle gates.
AIHE-R — VALIDATE-AI Implementation Plan Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Manage implementation as an evidence-producing sequence from problem verification through continuous evaluation.
AIHE-S — Algorithmovigilance Dashboard, Cadence, and Incident Record Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Operate daily technical checks, periodic service and economic review, incident classification, reconstruction, escalation, and annual renewal.
AIHE-T — AI De-adoption and Decommissioning Plan Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Define stopping triggers, fallback, transition, data disposition, communication, contract exit, validation, and post-retirement review.
AIHE-V — Validation Protocol for the Proposed Framework Package Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Define construct, content, reliability, feasibility, decision, implementation, and prospective validation for proposed frameworks.
AIHE-LTX-12-REF — Reference Resources nan; TeX	Chapter 12 reference resource aligned with the current book structure.
AIHE-LTX-12-SUP — Supplemental Resources nan; TeX	Chapter 12 supplemental resource aligned with the current book structure.
AIHE-LTX-12-WOR — Readmission-Risk Lifecycle Decision nan; TeX	Chapter 12 worked-example resource aligned with the current book structure.

Part V: Circular Economy and Sustainable Value

This Part applies resource stewardship, circular procurement, asset decisions, and circularity-adjusted evaluation to sustainable health-system value.

Chapter 13: Circular Health Systems and Resource Stewardship

Chapter Orientation

Chapter 13 extends stewardship to energy, water, computing, storage, hardware, materials, maintenance, sharing, recovery, resilience, and burden transfer. The resources apply a healthcare circularity hierarchy and distinguish lower resource intensity from lower absolute demand.

Use this suite when digital or AI decisions materially affect infrastructure or supply chains. It helps identify opportunities to avoid unnecessary acquisition, optimize demand, share capacity, extend safe useful life, document resource flows, test rebound, and build institutional circular operating capability.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Circular Stewardship of AI Compute and Clinical Infrastructure	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Circularity Hierarchy and Opportunity Register; Resource-Flow and Intensity Record; Circular Operating Capability Assessment	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Circular health system, Circularity hierarchy, Rebound effect	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	1 core, 2 python analytics, 1 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Circular Stewardship of AI Compute and Clinical Infrastructure

Purpose and Status

A health system is expanding AI-enabled imaging and analytics. Several servers are underutilized, cooling demand is increasing, equipment is approaching support milestones, and suppliers offer different maintenance, take-back, and product-information arrangements.

Decision Problem and Comparators

Decision problem. Which combination of demand prevention, shared use, maintenance, modernization, and recovery provides the greatest safe and sustainable value?

Table: Comparators for the Chapter 13 worked example

Option	Comparator or strategy
1	Continue current decentralized infrastructure
2	Consolidate and share existing capacity
3	Replace with new local infrastructure
4	Use a regional managed service
5	Reduce model and data demand before adding capacity

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$I_R = \frac{R}{Q}, R_{total} = I_R \times Q$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 13 worked example

Parameter	Base value	Source or interpretation
Average server utilization	28%	Infrastructure monitoring
Annual electricity use	1.8 GWh	Facilities data
Water use for cooling	Site estimate unavailable	Evidence gap
Assets within two years of support end	37%	Asset register
Suppliers offering take-back	2 of 5	Contract review
Potential model-efficiency reduction in compute/task	35%	Technical benchmark
Expected service growth	22% over three years	Portfolio forecast

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Define a clinically meaningful functional unit and map material, energy, water, data, and service flows.	Evidence and responsible owner to be recorded
2	Apply the circularity hierarchy before assuming that new acquisition is required.	Evidence and responsible owner to be recorded
3	Compare prevention, reduction, maintenance, sharing, reuse, refurbishment, replacement, and recovery options.	Evidence and responsible owner to be recorded
4	Assess whether AI-enabled forecasting or maintenance changes an actual resource decision.	Evidence and responsible owner to be recorded
5	Measure resource intensity and total demand to detect rebound.	Evidence and responsible owner to be recorded
6	Evaluate resilience, equity, supplier dependence, burden transfer, and final disposition.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Highest immediate opportunity	Consolidation and shared capacity	Low utilization indicates avoidable duplication
Efficiency effect	Potentially material	May be offset by service growth
Critical evidence gap	Cooling-water and embodied-impact data	Prevents complete comparison
Support risk	High for 37% of assets	Requires staged transition
Preferred decision	Demand reduction plus shared-capacity and supportability programme	Delay broad replacement

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Adopt a portfolio programme that first reduces unnecessary compute and storage, consolidates safe shared capacity, maintains supported assets, and uses replacement selectively where security, reliability, or efficiency justify it. Require supplier information, take-back, and traceable disposition.

- resource intensity and absolute demand are both monitored.
- safety, cybersecurity, resilience, and service continuity constrain every circular option.
- supplier and product data are sufficient for maintenance and recovery decisions.
- benefits and burdens beyond the hospital boundary are documented.
- future procurement includes durability, support, interoperability, take-back, and residual-value requirements.

Supplemental Resources

These supplemental resources translate the Chapter 13 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 13 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Circularity Hierarchy and Opportunity Register	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Resource-Flow and Intensity Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Circular Operating Capability Assessment	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Circularity Hierarchy and Opportunity Register

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Resource-Flow and Intensity Record

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define a clinically meaningful functional unit and map material, energy, water, data, and service flows. - Apply the circularity hierarchy before assuming that new acquisition is required. - Compare prevention, reduction, maintenance, sharing, reuse, refurbishment, replacement, and recovery options. - Assess whether AI-enabled forecasting or maintenance changes an actual resource decision. - Measure resource intensity and total demand to detect rebound. - Evaluate resilience, equity, supplier dependence, burden transfer, and final disposition.

Circular Operating Capability Assessment

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 13

Term	Working definition and decision relevance
Circular health system	A health system that preserves health and public value while reducing unnecessary resource demand and retaining useful value.
Circularity hierarchy	A decision order prioritizing prevention and higher-value retention before recycling and disposal.
Rebound effect	Growth in activity that offsets efficiency gains and increases total resource demand.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 13

Instrument	Status or type	Appropriate use and limitation
ISO 59004	Voluntary circular-economy standard	Vocabulary, principles, and implementation guidance.
ISO 59010	Voluntary circular-economy standard	Guidance on value-network transition.
ISO 59020	Voluntary circularity-measurement standard	Measurement and assessment of circularity performance.
ISO 59014/59040	Voluntary standards	Traceability of secondary materials and product circularity information.

Formula and Interpretation Note

The following relationship is used in Chapter 13 or its companion resources:

$$I_R = \frac{R}{Q}, R_{total} = I_R \times Q$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE13 — Circular AI Infrastructure Strategy Worked case; md; csv; json; xlsx; ods	Can the health system meet its analytic need through avoidance, optimization, sharing, and life extension before acquiring new infrastructure?
AIHE-E01 — Circular-Economy and Sustainable-AI Assessment Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Apply the circularity hierarchy from prevention and reduction through sharing, maintenance, reuse, refurbishment, recovery, and disposal.
AIHE-PY36 — Environmental, material-flow, and lifecycle resource accounting Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Quantifies lifecycle cost, energy, water, carbon, cost per useful output, and circularity indicators for AI-related assets and services.
AIHE-PY39 — Annual circularity portfolio review and trend monitoring Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Provides a reproducible synthetic example for annual circularity portfolio review and trend monitoring.
AIHE-CE1 — Circular Economy and Sustainable AI Assessment Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Assess service need, the circularity hierarchy, resource demand, rebound, resilience, equity, evidence, and lifecycle review across AI-enabled health services.
AIHE-LTX-13-REF — Reference Resources nan; TeX	Chapter 13 reference resource aligned with the current book structure.
AIHE-LTX-13-SUP — Supplemental Resources nan; TeX	Chapter 13 supplemental resource aligned with the current book

Resource	Purpose
	structure.
AIHE-LTX-13-WOR — Circular Stewardship of AI Compute and Clinical Infrastructure nan; TeX	Chapter 13 worked-example resource aligned with the current book structure.

Chapter 14: Circular Procurement and Asset Stewardship

Chapter Orientation

Chapter 14 translates circular principles into procurement and asset decisions. The resources compare maintain, repair, refurbish, remanufacture, replace, share, recover, and decommission options while preserving safety, supportability, cybersecurity, availability, data closure, and continuity.

Use this suite for equipment and platform refresh decisions, contract specifications, maintenance planning, residual-value assessment, and controlled retirement. The imaging-platform case demonstrates why useful-life extension is a conditional option rather than an objective in itself.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Imaging Platform Refresh: Maintain, Refurbish, Replace, or Share	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Asset Option Appraisal; Maintenance and Supportability Record; Decommissioning and Disposition Record	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Asset stewardship, Refurbishment, Responsible decommissioning	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	2 core, 1 python analytics, 2 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Imaging Platform Refresh: Maintain, Refurbish, Replace, or Share

Purpose and Status

A hospital imaging platform depends on ageing local servers, proprietary interfaces, and a supported but approaching end-of-life software stack. The hospital must compare continued maintenance, modular refurbishment, full replacement, and a shared managed service.

Decision Problem and Comparators

Decision problem. Which asset strategy preserves safe service, value, resilience, and exit capability over the next five years?

Table: Comparators for the Chapter 14 worked example

Option	Comparator or strategy
1	Maintain current platform for a limited period
2	Modular refurbishment and software modernization
3	Full local replacement
4	Regional shared managed service

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$EAC_j = PV(C_j)\left[\frac{r(1+r)^{n_j}}{(1+r)^{n_j}-1}\right)$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 14 worked example

Parameter	Base value	Source or interpretation
Current asset age	7 years	Asset register
Remaining vendor support	24 months	Support contract
Unplanned downtime	38 hours/year	Service records
Five-year PV cost: maintain	EUR 2.4 million	Option model
Five-year PV cost: refurbish	EUR 3.1 million	Option model
Five-year PV cost: replace	EUR 4.6 million	Option model
Five-year PV cost: shared service	EUR 3.8 million	Option model
Estimated recoverable residual value	EUR 280,000	Market and take-back estimate

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Define functional service requirements and non-negotiable safety, support, cybersecurity, and continuity conditions.	Evidence and responsible owner to be recorded
2	Compare lifecycle cost and equivalent annual cost only among eligible options.	Evidence and responsible owner to be recorded
3	Assess compatibility, validation burden, downtime, maintainability, software support, energy, residual value, and exit.	Evidence and responsible owner to be recorded
4	Examine reuse, refurbishment, transfer, take-back, secure erasure, and chain of custody.	Evidence and responsible owner to be recorded
5	Test downside scenarios for failure, vendor exit, delayed migration, and demand growth.	Evidence and responsible owner to be recorded
6	Document the transition, data closure, receiving party, final destination, and organizational learning.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Maintain option	Lowest apparent cost	Fails supportability beyond 24 months without a transition
Refurbish option	Strong short-term value	Depends on software and security support
Replace option	Highest initial cost	Best control but substantial embodied resource and transition burden
Shared-service option	Moderate cost and resilience potential	Data, dependency, and exit terms are decision critical

Output	Result	Interpretation
Preferred strategy	Modular refurbishment with prepared shared-service option	Preserves flexibility

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Undertake modular refurbishment to preserve safe useful life while preparing a contractually robust shared-service option. Do not extend the current platform beyond supported and validated operation. Use recovery value and take-back only where chain of custody and final disposition are verified.

- support, cybersecurity, and validation remain acceptable throughout the planned period.
- equivalent annual cost uses a common boundary and realistic residual value.
- data migration, secure erasure, and fallback are tested before transition.
- spare parts, documentation, and specialist capability are available.
- renewal and replacement decisions are reviewed before the support deadline.

Supplemental Resources

These supplemental resources translate the Chapter 14 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 14 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Asset Option Appraisal	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Maintenance and Supportability Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Decommissioning and Disposition Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Asset Option Appraisal

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Maintenance and Supportability Record

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define functional service requirements and non-negotiable safety, support, cybersecurity, and continuity conditions. - Compare lifecycle cost and equivalent annual cost only among eligible options. - Assess compatibility, validation burden, downtime, maintainability, software support, energy, residual value, and exit. - Examine reuse, refurbishment, transfer, take-back, secure erasure, and chain of custody. - Test downside scenarios for failure, vendor exit, delayed migration, and demand growth. - Document the transition, data closure, receiving party, final destination, and organizational learning.

Decommissioning and Disposition Record

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 14

Term	Working definition and decision relevance
Asset stewardship	Responsible management of performance, support, useful life, value, risk, and disposition.
Refurbishment	Restoration to a specified condition through controlled inspection, repair, replacement, testing, and documentation.
Responsible decommissioning	Controlled service transition, data closure, recovery, transfer, and final disposition.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 14

Instrument	Status or type	Appropriate use and limitation
ISO 55001	Voluntary asset-management standard	Management system for value, performance, risk, expenditure, and lifecycle control.
Ecodesign for Sustainable Products Regulation	Binding EU regulation in scope	Framework for ecodesign and product-information requirements.
Global E-waste Monitor	International evidence resource	Documents e-waste generation, formal collection, and recovery.
Circular procurement	Procurement approach	Embeds durability, support, repair, recovery, and exit in specifications and contracts.

Formula and Interpretation Note

The following relationship is used in Chapter 14 or its companion resources:

$$EAC_j = PV(C_j)\left[\frac{r(1+r)^{n_j}}{(1+r)^{n_j}-1}\right]$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE14 — Imaging Platform: Maintain, Refurbish, Replace, or Share Worked case; md; csv; json; xlsx; ods	Which asset-life strategy preserves safe service value while controlling lifecycle cost, downtime, material demand, and dependency?
AIHE-E02 — Circular Procurement and Supplier Scorecard Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Evaluate durability, modularity, repairability, interoperability, resource information, take-back, pricing, residual value, and exit.
AIHE-E03 — Asset-Life Decision Record Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Compare maintain, refurbish, replace, and shared-service options across suitability, life, cost, resources, resilience, and disposition.
AIHE-PY37 — Circular asset-life option analysis Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Ranks circular asset and service options using discounted cost, carbon, downtime, circularity, safety, and non-compensatory safeguards.
AIHE-CE2 — Circular Procurement and Supplier Scorecard Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Evaluate lifecycle specifications, durability, modularity, repairability, service support, product circularity data, take-back, residual value, and responsible end of life.
AIHE-CE3 — Circular Asset-Life Decision Workbook Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Compare maintain, repair, upgrade, share, refurbish, remanufacture, replace, and service-model options using safety, cost, life, validation,

Resource	Purpose
	resilience, and residual value.
AIHE-LTX-14-REF — Reference Resources nan; TeX	Chapter 14 reference resource aligned with the current book structure.
AIHE-LTX-14-SUP — Supplemental Resources nan; TeX	Chapter 14 supplemental resource aligned with the current book structure.
AIHE-LTX-14-WOR — Imaging Platform Refresh: Maintain, Refurbish, Replace, or Share nan; TeX	Chapter 14 worked-example resource aligned with the current book structure.

Chapter 15: Circularity-Adjusted Evaluation and Sustainable Value

Chapter Orientation

Chapter 15 extends conventional economic evaluation through explicit environmental, material, residual-value, resilience, distributional, and rebound evidence. The resources preserve conventional results while showing where physical indicators, monetized adjustments, and multi-criteria deliberation should remain separate.

Use this suite when architecture, infrastructure, or asset options differ materially in energy, water, hardware, recovery, resilience, or burden distribution. The edge-local-cloud case demonstrates how the preferred decision can change when system boundaries and functional units are made explicit.

Resource Suite at a Glance

Layer	Chapter resource	How to use it
Worked example	Worked Example: Edge, Local, or Cloud Diagnostic AI	Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions.
Supplemental resources	Circularity-Adjusted Evaluation Protocol; Environmental and Material Evidence Record; Multi-Criteria Deliberation and Safeguards	Create the institutional records needed for analysis, approval, monitoring, and review.
Reference focus	Functional unit, Circularity-adjusted value, Sustainable value	Clarify terminology, method selection, instrument status, formulas, and update requirements.
Repository coverage	2 core, 1 python analytics, 4 supplemental specialist tool, 1 worked case	Use the catalog map below to locate editable, process-flow, data, and computational variants.

Worked Example: Edge, Local, or Cloud Diagnostic AI

Purpose and Status

A diagnostic service is comparing three deployment architectures: upgrade existing edge infrastructure, replace it with efficient local infrastructure, or use a shared cloud service. Clinical performance is assumed equivalent, but lifecycle cost, energy, water, hardware, data transfer, resilience, residual value, and exit differ.

Decision Problem and Comparators

Decision problem. Which architecture offers the strongest sustainable value after conventional and circularity-related effects are considered?

Table: Comparators for the Chapter 15 worked example

Option	Comparator or strategy
1	Upgrade existing edge servers
2	Replace with new local infrastructure
3	Shared cloud service
4	Reduce model size and data retention before selecting architecture

Model and Decision Structure

A compact quantitative relationship used in this example is:

$$CANB_j = \lambda \Delta E_j - \Delta C_j - \Delta EC_j + \Delta RV_j + \Delta SRV_j$$

Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 15 worked example

Parameter	Base value	Source or interpretation
Annual validated analyses	1,200,000	Functional-unit denominator
Five-year PV cost: edge upgrade	EUR 3.2 million	Lifecycle model
Five-year PV cost: local replacement	EUR 4.1 million	Lifecycle model
Five-year PV cost: cloud	EUR 3.7 million	Lifecycle model
Operational energy per 1,000 analyses	Edge 42; local 27; cloud allocation 19 kWh	Illustrative inputs
Expected hardware residual value	Edge EUR 70,000; local EUR 240,000	Recovery estimate
Maximum tolerable service interruption	30 minutes	Clinical continuity requirement
Annual demand growth	18%	Scenario assumption

Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

Step	Required work	Decision record
1	Define the functional unit, perspective, system boundary, horizon, and comparators.	Evidence and responsible owner to be recorded
2	Combine lifecycle costing with physical energy, water, material, residual-value, and resilience evidence.	Evidence and responsible owner to be recorded
3	Report impacts separately where monetization is not credible.	Evidence and responsible owner to be recorded
4	Estimate resource intensity and absolute demand, including rebound under growth scenarios.	Evidence and responsible owner to be recorded
5	Use circularity-adjusted net benefit only with consistent valuation, no double counting, and explicit limitations.	Evidence and responsible owner to be recorded
6	Apply multi-criteria deliberation after mandatory safety, privacy, cybersecurity, continuity, and equity safeguards are met.	Evidence and responsible owner to be recorded

Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

Output	Result	Interpretation
Lowest operational energy intensity	Cloud under the stated allocation	Allocation and location uncertainty are material
Lowest five-year PV cost	Edge upgrade	Support and resilience limits may offset cost advantage
Best local control and residual value	Local replacement	Higher initial and embodied resource burden
Strongest scalability	Cloud	Vendor dependence, data transfer, and exit require controls
Preferred decision	Optimize demand, then use a hybrid local/shared architecture	Robust across uncertainty

Sensitivity, Uncertainty, and Subgroups

Interpretation and Recommendation

Reduce model and data demand first, retain local capacity for time-critical and continuity-sensitive functions, and use shared infrastructure where allocation, security, location, portability, and exit evidence are acceptable. Report monetary and physical outcomes separately and revisit the architecture as demand and technology change.

- clinical effectiveness is comparable across architectures.
- the functional unit and allocation rules are documented.
- absolute demand and rebound are monitored, not only efficiency per task.
- resilience, privacy, cybersecurity, and exit are non-compensatory.
- the decision record reports which effects are monetized, disaggregated, or considered through deliberation.

Supplemental Resources

These supplemental resources translate the Chapter 15 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 15 supplemental-resource set

Resource	Purpose and minimum content	Expected output
Circularity-Adjusted Evaluation Protocol	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Environmental and Material Evidence Record	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.
Multi-Criteria Deliberation and Safeguards	Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action.	Blank record, synthetic example, and version history.

Circularity-Adjusted Evaluation Protocol

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

Environmental and Material Evidence Record

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define the functional unit, perspective, system boundary, horizon, and comparators. - Combine lifecycle costing with physical energy, water, material, residual-value, and resilience evidence. - Report impacts separately where monetization is not credible. - Estimate resource intensity and absolute demand, including rebound under growth scenarios. - Use circularity-adjusted net benefit only with consistent valuation, no double counting, and explicit limitations. - Apply multi-criteria deliberation after mandatory safety, privacy, cybersecurity, continuity, and equity safeguards are met.

Multi-Criteria Deliberation and Safeguards

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

Reference Resources

Key Terms

Table: Key terms for Chapter 15

Term	Working definition and decision relevance
Functional unit	The defined service output to which costs and environmental or material effects are related.
Circularity-adjusted value	Conventional economic value extended by explicit circularity, residual-value, resilience, and environmental evidence.
Sustainable value	Long-term health-system value incorporating affordability, resilience, equity, environmental effects, and public purpose.

Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 15

Instrument	Status or type	Appropriate use and limitation
Life-cycle assessment	Environmental method	Quantifies environmental impacts across a defined life cycle.
Material-flow analysis	Physical accounting method	Tracks material stocks and flows.
Life-cycle costing	Economic method	Estimates costs across acquisition, use, transition, and exit.
ISO 59020	Circularity-measurement standard	Supports defined boundaries, indicators, data, and interpretation.
Multi-criteria decision analysis	Decision method	Structures deliberation when effects cannot be credibly monetized.

Formula and Interpretation Note

The following relationship is used in Chapter 15 or its companion resources:

$$CANB_j = \lambda \Delta E_j - \Delta C_j - \Delta EC_j + \Delta RV_j + \Delta SRV_j$$

Its variables, perspective, time horizon, uncertainty, and limitations must be defined before use. A formula should not be added merely for symmetry with other chapters; it is useful only when it clarifies a decision and can be populated with defensible evidence.

Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

Resource	Purpose
AIHE-CASE15 — Edge, Cloud, or Shared Diagnostic AI: Circularity-Adjusted Evaluation Worked case; md; csv; json; xlsx; ods	How does the preferred option change when conventional economic value is considered with environmental effects, residual value, resilience, equity, and rebound?
AIHE-E04 — Circularity-Adjusted Evaluation Record Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Extend conventional economic evaluation with environmental, material, residual-value, resilience, equity, rebound, and burden evidence.
AIHE-E05 — Environmental, Circularity, and Disposition Dashboard Core; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Monitor acquisition, utilization, asset life, resource intensity, materials, take-back, recovery, rebound, resilience, and disposition.
AIHE-PY38 — Circularity-adjusted economic evaluation and sustainable value Python analytics; py; ipynb; xlsx; ods; csv; json; png; md	Provides a reproducible synthetic example for circularity-adjusted economic evaluation and sustainable value.
AIHE-CE4 — Environmental Footprint and Circularity Dashboard Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Monitor absolute and per-functional-unit energy, water, carbon, materials, utilization, repair, life extension, recovery, rebound, resilience, and distributional effects.
AIHE-CE5 — Circular Decommissioning and Asset Disposition Record Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Document continuity, secure erasure, chain of custody, recipient due diligence, reuse, refurbishment, recovery, residual value, communication, and closure.
AIHE-CE6 — Circularity-Adjusted Evaluation Record Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Specify a common functional unit and boundary; integrate clinical, lifecycle-cost, environmental, residual-value, resilience, equity, rebound, and uncertainty evidence without arbitrary monetization or double counting.

Resource	Purpose
AIHE-CE7 — Annual Circularity Review Supplemental specialist tool; docx; odt; xlsx; ods; csv; json; md; mmd; svg; png	Conduct a portfolio-level annual review of resource flows, asset-life extension, environmental trends, supplier performance, recovery, rebound, incidents, capability gaps, and next-year decisions.
AIHE-LTX-15-REF — Reference Resources nan; TeX	Chapter 15 reference resource aligned with the current book structure.
AIHE-LTX-15-SUP — Supplemental Resources nan; TeX	Chapter 15 supplemental resource aligned with the current book structure.
AIHE-LTX-15-WOR — Edge, Local, or Cloud Diagnostic AI nan; TeX	Chapter 15 worked-example resource aligned with the current book structure.

Cross-Chapter and Optional Resources

The resources below apply across several chapters or support teaching, case analysis, and publication maintenance. They should be used selectively and kept separate from the core institutional decision record unless they directly inform the decision.

Resource	Tier	Purpose	Formats
AIHE-CASE00 — Common Case-Analysis Method	Case-analysis method	Provide a consistent problem-first, evidence, economic, governance, and lifecycle method for analysing worked and local cases.	md; docx; odt; xlsx; ods; csv; json
AIHE-AG — Editable Figure and Graph Source Manifest	Optional reference	Record figure files, data, code, ownership, accessibility, licensing, version, and update status.	docx; odt; xlsx; ods; csv; json; md
AIHE-AD — Fourteen-Week Course Schedule	Optional teaching	Provide an optional semester sequence for instructors using the book and companion resources.	docx; odt; xlsx; ods; csv; json; md
AIHE-F — Suggested Twelve-Week Teaching Plan	Optional teaching	Provide a compressed optional teaching sequence separate from the institutional decision records.	docx; odt; xlsx; ods; csv; json; md

Repository Workflow for Institutional Projects

Suggested Project Folder

```
project-name/
├── 00_source_release/        # Unmodified companion files and citation record
├── 01_decision_specification/ # Problem, comparator, intended use, owners
├── 02_evidence/              # External and local evidence, appraisal, versions
├── 03_analysis/              # Costing, models, notebooks, tests, outputs
├── 04_governance/            # Safeguards, approvals, risk, contracts
├── 05_implementation/        # Training, workflow, pilot, monitoring
├── 06_decisions/             # Signed decisions, conditions, dissent, review dates
└── 07_closure/               # Transition, data closure, decommissioning, lessons
```

Control Principles

- Keep the published release unchanged and work from a controlled copy.
- Use one accountable document owner and named owners for specialist analyses.
- Link each calculation and statement to a source, date, and version.
- Archive static outputs from interactive notebooks and dashboards when they inform a decision.
- Retain rejected alternatives and dissent so that later review can reconstruct the original choice.
- Use event-triggered review after incidents, material changes, vendor notices, major price changes, new evidence, altered populations, or changed workflows.
- Close a project only after continuity, communication, data retention or deletion, contract closure, asset disposition, and organizational learning are documented.

Customization and Quality Review

A local adaptation should identify what changed and why. Edits that alter a definition, threshold, formula, score, mandatory safeguard, or decision rule require more scrutiny than edits to layout or local terminology. Before approval, reviewers should confirm that:

1. the decision problem and comparator remain clear;
2. the exact intervention and version are traceable;
3. the data and evidence are current and applicable;
4. assumptions and missing evidence are visible;
5. calculations are reproducible and independently checked in proportion to consequence;
6. subgroup, workload, affordability, continuity, and exit effects are not omitted;
7. mandatory safeguards are assessed independently of any aggregate score;
8. the decision conditions and monitoring triggers are operational; and
9. the completed record is stored, cited, and scheduled for review.

Versioning and Citation

A recommended citation structure is:

> Zouri, M., and Cumpăt, C. M. Artificial Intelligence and Healthcare Economics: Digital Resource Companion. V1.0. 2026. Cite the DOI and repository URL associated with the specific release used.

For a specific resource, add the stable identifier and title:

> AIHE-[ID], [Resource title], V1.0, in the Digital Resource Companion, accessed [date], locally adapted by [institution] on [date].

A modified resource should retain the original citation and add a local version statement. Do not imply that a local adaptation has been reviewed or endorsed by the authors unless such review has occurred.

Glossary

Term	Definition
AI Health Economics Dossier (AI-HED)	A versioned evidence package linking the decision problem, evidence, economics, governance, implementation, monitoring, material change, and final recommendation
AI Health-System Value Chain	A proposed causal sequence extending from problem and data through model output, decision, action, outcome, resource consequence, and public value
AI-HEVG	The proposed Artificial Intelligence Health Economic Value and Governance Framework, integrating problem relevance, data readiness, clinical validity, economic value, ethical acceptability, organizational feasibility, and lifecycle governance
algorithmic bias	Systematic error or unfairness in artificial intelligence outputs or pathways that disadvantages particular groups, settings, or circumstances
algorithmovigilance	Continuing surveillance and governance of the real-world technical, clinical, economic, ethical, equity, workflow, cybersecurity, and trust performance of artificial intelligence-enabled interventions
artificial intelligence	A broad family of computational techniques designed to perform tasks associated with human intelligence, including pattern recognition, prediction, language processing, classification, generation, optimization, and decision support
budget impact analysis	An assessment of the expected financial consequences of adopting an intervention within a specified budget, population, uptake pattern, and time horizon
circular economy	A systems approach that seeks to preserve the value of products, components, materials, infrastructure, and services while reducing virgin-resource use, waste, and environmental burden
circular health system	A health system that applies prevention, reduction, maintenance, sharing, reuse, repair, refurbishment, remanufacture, recovery, and responsible disposition while protecting clinical quality, safety, equity, and resilience
circular procurement	Procurement that evaluates the service need and full lifecycle, including durability, maintenance, repair, interoperability, useful life, environmental information, take-back, residual value, and exit
circularity-adjusted net benefit	An author-developed scenario-analysis expression extending conventional net monetary benefit with selected monetized environmental, residual-value, and resilience terms while retaining disaggregated outcomes
clinical decision support	Tools that provide patient-specific information, predictions, alerts, or recommendations to support clinical decisions
corrective and preventive action	A structured process for containing a problem, investigating causes, implementing corrective and preventive measures, and verifying their effectiveness
cost-effectiveness analysis	An economic evaluation comparing the incremental costs and outcomes of alternative interventions
cost-utility analysis	A form of cost-effectiveness analysis in which outcomes are commonly expressed as quality-adjusted life years
data provenance	Documented information about the origin, ownership, timing, transformation, version, and lineage of data used in an analysis or artificial intelligence system
de-adoption	A deliberate reduction, restriction, replacement, or cessation of an intervention that no longer provides an acceptable balance of value, evidence, feasibility, and risk
decommissioning	The planned technical, contractual, operational, data, and asset activities required to retire a digital or artificial intelligence system safely
digital exclusion	Barriers that prevent people from benefiting from digital health services because of access, literacy, language, disability, income, geography, design, or trust
digital product passport	A digital information mechanism used in European sustainable-product policy to provide product information across the value chain, subject to

Term	Definition
	product-specific requirements
EWAHE	The proposed Equity-Weighted AI Health Economics process for examining subgroup performance, access, outcomes, costs, distribution, mitigation, and revised decisions
GATED-AI	A proposed sequential procurement and approval process covering problem, evidence, data, economics, ethics, workflow, contract, monitoring, and exit gates
generative AI	Artificial intelligence systems that generate text, images, code, summaries, recommendations, or other outputs in response to prompts or contextual inputs
health technology assessment	A multidisciplinary process that evaluates the clinical, economic, organizational, ethical, and social implications of a health technology to support policy and adoption decisions
HOAM	The proposed Human Oversight and Accountability Matrix, which assigns accountable, responsible, consulted, and informed roles across the artificial intelligence lifecycle
human oversight	Operational arrangements that enable people to understand, supervise, question, intervene in, override, restrict, or stop an artificial intelligence system
interoperability	The ability of systems and organizations to exchange, interpret, and use information consistently across technical, semantic, organizational, and temporal boundaries
joint clinical assessment	A collaborative European Union health technology assessment of relative clinical effectiveness and safety under Regulation (EU) 2021/2282, while national authorities retain responsibility for economic appraisal and reimbursement decisions
life-cycle assessment	A method for assessing environmental burdens across defined stages of a product or service lifecycle, from resource extraction through use and end of life
life-cycle costing	An approach that estimates costs across acquisition, implementation, use, maintenance, updating, transition, and end-of-life stages within a defined boundary
local validation	Evaluation of a system using data, users, workflows, populations, and operational conditions from the intended local setting before or during controlled deployment
machine learning	A subset of artificial intelligence in which a model learns patterns or decision functions from data rather than relying only on explicitly programmed rules
material change	A change in a model, software, data, threshold, intended use, population, workflow, infrastructure, price, vendor arrangement, or governance that may alter performance, risk, value, or obligations
material-flow analysis	A method for tracing the quantities and movement of products, components, or materials into, within, and out of a defined system
model drift	A decline or change in model performance over time because data, populations, prevalence, workflow, coding, practice, or user behavior have changed
net monetary benefit	A decision measure that values incremental outcomes at a stated threshold and subtracts incremental cost, facilitating comparison under uncertainty
non-compensatory safeguard	A minimum requirement that cannot be offset by favorable scores in other domains, such as patient safety, lawful data use, privacy, cybersecurity, equity, continuity, or accountability
product circularity data sheet	A standardized structure for communicating product information relevant to circularity, value retention, recovery, and lifecycle decisions
Public Value AI Scorecard	A proposed structured summary of clinical, economic, equity, workforce, patient, governance, strategic, and sustainability value
quality-adjusted life year	A measure combining duration and health-related quality of life, with one unit representing one year in full health

Term	Definition
rebound effect	An increase in total resource use, demand, or downstream activity that partly or fully offsets an efficiency improvement
repairability	The practical ability to restore function through accessible parts, documentation, diagnostics, skills, service rights, software support, and economically feasible repair
residual value	The realizable value remaining in an asset, component, material, licence, or service arrangement at a specified point, net of costs required to recover that value
return on investment	A financial measure comparing realized gains or savings with costs; in health systems it should be interpreted alongside clinical, distributional, organizational, and public value
SHARE-AI	A proposed process for specifying purpose, harmonizing data, authorizing access, reviewing risks, evaluating value, and auditing implementation of secondary data use
statistical process control	The use of time-ordered data and control limits to distinguish common-cause variation from signals that warrant investigation or corrective action
TAV	The proposed Trust-Adjusted Value framework for interpreting clinical and economic value in light of transparency, fairness, patient trust, professional trust, and public accountability
TCAIO	The proposed Total Cost of AI Ownership framework, including acquisition, integration, data, workforce, governance, cybersecurity, monitoring, opportunity, material-change, and exit costs
technical debt	The future cost and risk created when short-term technical choices leave fragile interfaces, undocumented transformations, outdated dependencies, or deferred maintenance
total cost of ownership	The full lifecycle cost of acquisition, implementation, integration, operation, monitoring, updating, support, transition, and exit
VALIDATE-AI	A proposed implementation pathway comprising Verify, Assess, Local validation, Implement pilot, Determine value, Adapt governance, Transition to scale, and Evaluate continuously
value of information	An analytic approach estimating the expected benefit of reducing decision uncertainty through additional evidence

Abbreviations and Acronyms

Abbreviation	Meaning
ABM	agent-based model
AI	artificial intelligence
AI-HED	AI Health Economics Dossier
AI-HEVG	Artificial Intelligence Health Economic Value and Governance Framework
API	application programming interface
BIA	budget impact analysis
CAPA	corrective and preventive action
CEA	cost-effectiveness analysis
CFIR	Consolidated Framework for Implementation Research
CHEERS-AI	Consolidated Health Economic Evaluation Reporting Standards for interventions using artificial intelligence
CONSORT-AI	Consolidated Standards of Reporting Trials–Artificial Intelligence extension
CUA	cost-utility analysis
DECIDE-AI	Developmental and Exploratory Clinical Investigations of Decision support systems driven by Artificial Intelligence
DES	discrete-event simulation
DPP	digital product passport
EHDS	European Health Data Space
EHR	electronic health record
EU	European Union
EVPI	expected value of perfect information
EVPPI	expected value of partial perfect information
EWAHE	Equity-Weighted AI Health Economics
FHIR	Fast Healthcare Interoperability Resources
GRADE-CERQual	Grading of Recommendations Assessment, Development and Evaluation–Confidence in the Evidence from Reviews of Qualitative research
HOAM	Human Oversight and Accountability Matrix
HTA	health technology assessment
ICER	incremental cost-effectiveness ratio
ISO/IEC	International Organization for Standardization and International Electrotechnical Commission
JCA	joint clinical assessment
LCA	life-cycle assessment
LCC	life-cycle costing
MFA	material-flow analysis
MLOps	machine-learning operations
NASSS	Nonadoption, Abandonment, Scale-up, Spread, and Sustainability
NICE	National Institute for Health and Care Excellence

Abbreviation	Meaning
NIST	National Institute of Standards and Technology
NMB	net monetary benefit
OECD	Organisation for Economic Co-operation and Development
OMOP	Observational Medical Outcomes Partnership common data model
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PROBAST+AI	Prediction model Risk Of Bias ASsessment Tool plus artificial intelligence
QALY	quality-adjusted life year
ROI	return on investment
SHARE-AI	Specify, Harmonize, Authorize, Review, Evaluate, and Audit
SPC	statistical process control
SPIRIT-AI	Standard Protocol Items: Recommendations for Interventional Trials–Artificial Intelligence extension
STARD-AI	Standards for Reporting Diagnostic Accuracy Studies–Artificial Intelligence extension
TAV	Trust-Adjusted Value
TCAIO	Total Cost of AI Ownership
TCO	total cost of ownership
TRIPOD+AI	Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis plus artificial intelligence
VALIDATE-AI	Verify, Assess, Local validation, Implement pilot, Determine value, Adapt governance, Transition to scale, and Evaluate continuously
WHO	World Health Organization

Responsible Use

These resources support structured analysis, documentation, education, and institutional deliberation. They do not replace professional judgement or the governance duties of the adopting organization. A completed template does not establish that an intervention is safe, lawful, effective, equitable, affordable, sustainable, or appropriate. The final responsibility is to make evidence, uncertainty, value judgements, ownership, safeguards, and the ability to change course visible enough for an accountable decision.

References

1. Hill H, Roadevin C. Economic evaluation of artificial intelligence for cancer detection in the UK breast screening programme. British Journal of Cancer. 2026;135:453–60.

2. Degeling K, Karnon J, Ven M van de, Brennan A, Koffijberg H. Discrete event simulation in R using the 'simmer' package for health economic modelling: A tutorial and illustration in colon cancer. Applied Health Economics and Health Policy. 2025;23(6):961–75.

3. Elvidge J, Hawksworth C, Avşar TS, Zemplenyi A, Chalkidou A, Petrou S, et al. Consolidated Health Economic Evaluation Reporting Standards for interventions that use artificial intelligence (CHEERS-AI). Value in Health. 2024;27(9):1196–205.

4. Kunst N, Siu A, Drummond M, Grimm S, Grutters J, Husereau D, et al. Reporting economic evaluations with value of information analyses using the CHEERS value of information (CHEERS-VOI) reporting guideline. Medical Decision Making. 2024;44(2):127–8.

5. Erdemir A, Mulugeta L, Ku JP, Drach A, Horner M, Morrison TM, et al. Credible practice of modeling and simulation in healthcare: Ten rules from a multidisciplinary perspective. Journal of Translational Medicine. 2020;18.